UVA CS 4774: Machine Learning

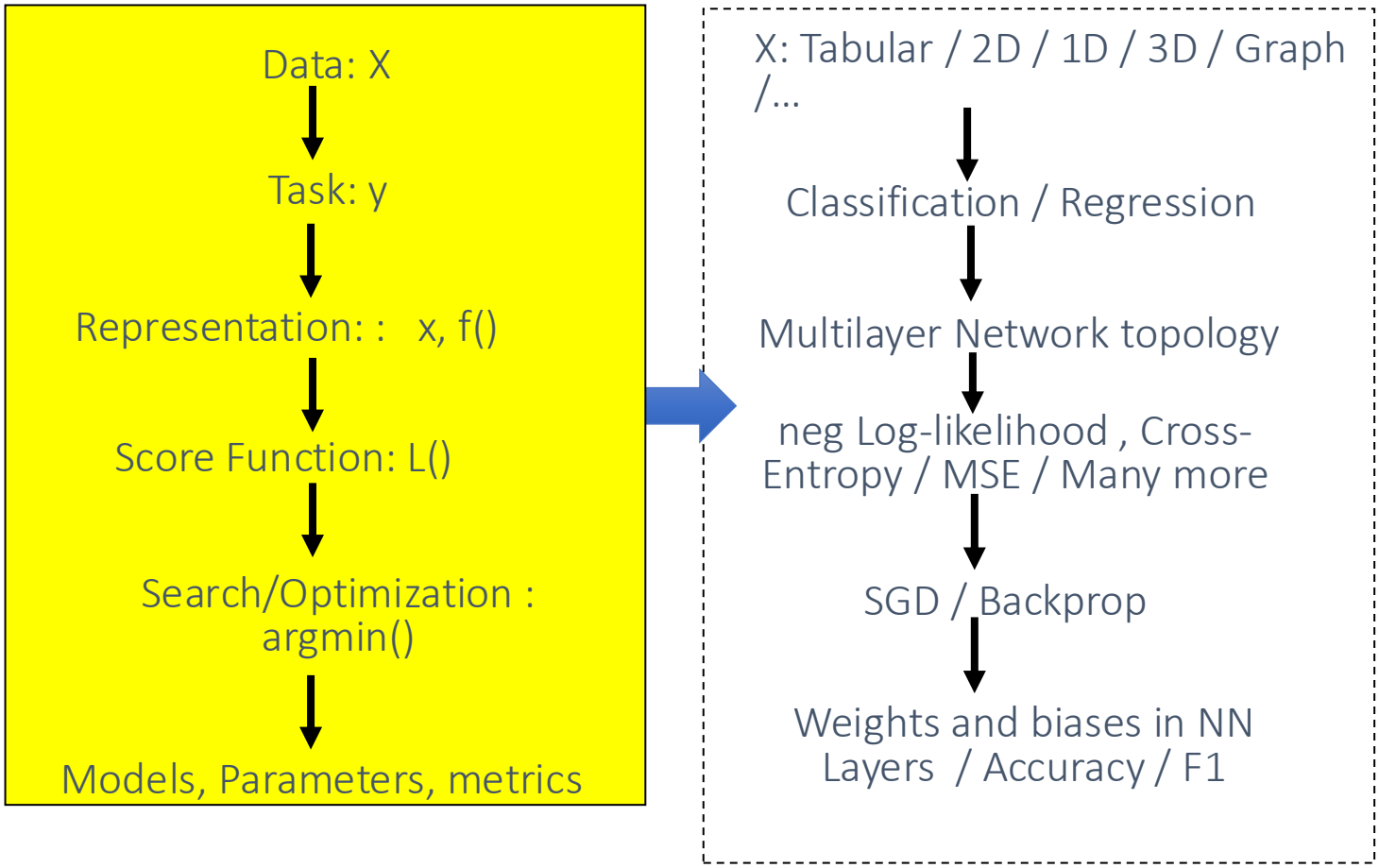
Lecture 13: Supervised Image Classification and Convolutional Neural Networks

Dr. Yanjun Qi

University of Virginia

Department of Computer Science

Last: Basic Neural Network Models



Early History

- In 1950 English mathematician Alan Turing wrote a landmark paper titled "Computing Machinery and Intelligence" that asked the question: "Can machines think?"
- Further work came out of a 1956 workshop at Dartmouth sponsored by John McCarthy. In the proposal for that workshop, he coined the phrase a "study of artificial intelligence"
- 1950s
  - Samuel's checker player : start of machine learning
  - Selfridge's Pandemonium
- 1952-1969: Enthusiasm: Lots of work on neural networks
- 1970s-80s: Expert systems, Knowledge bases to add on rule-based inference, Decision Trees, Bayes Nets
- 1990s : CNN, RNN, ….
- 2000s : SVM, Kernel machines, Structured learning, Graphical models, semi-supervised, matrix factorization, …

Adapted From Prof. Raymond J. Mooney's slides

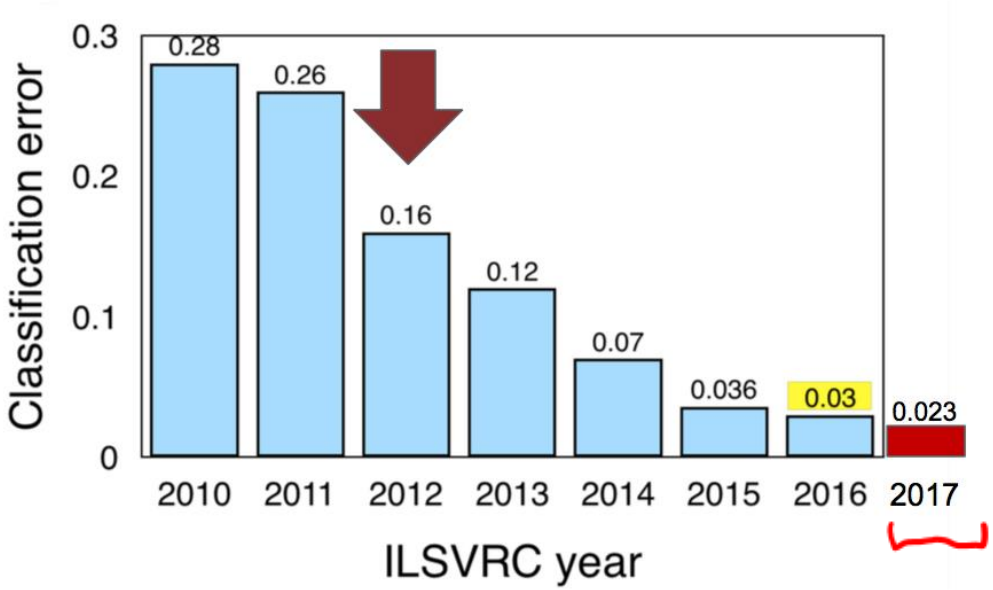
“Winter of Neural Networks” in ~2000s

- Non-convex
- Need a lot of tricks to play with
  - How many layers ?
  - How many hidden units per layer ?
  - What topology among layers ? …….
- Hard to perform theoretical analysis
- Large labeled datasets were rare in ~2000s

ImageNet Challenge

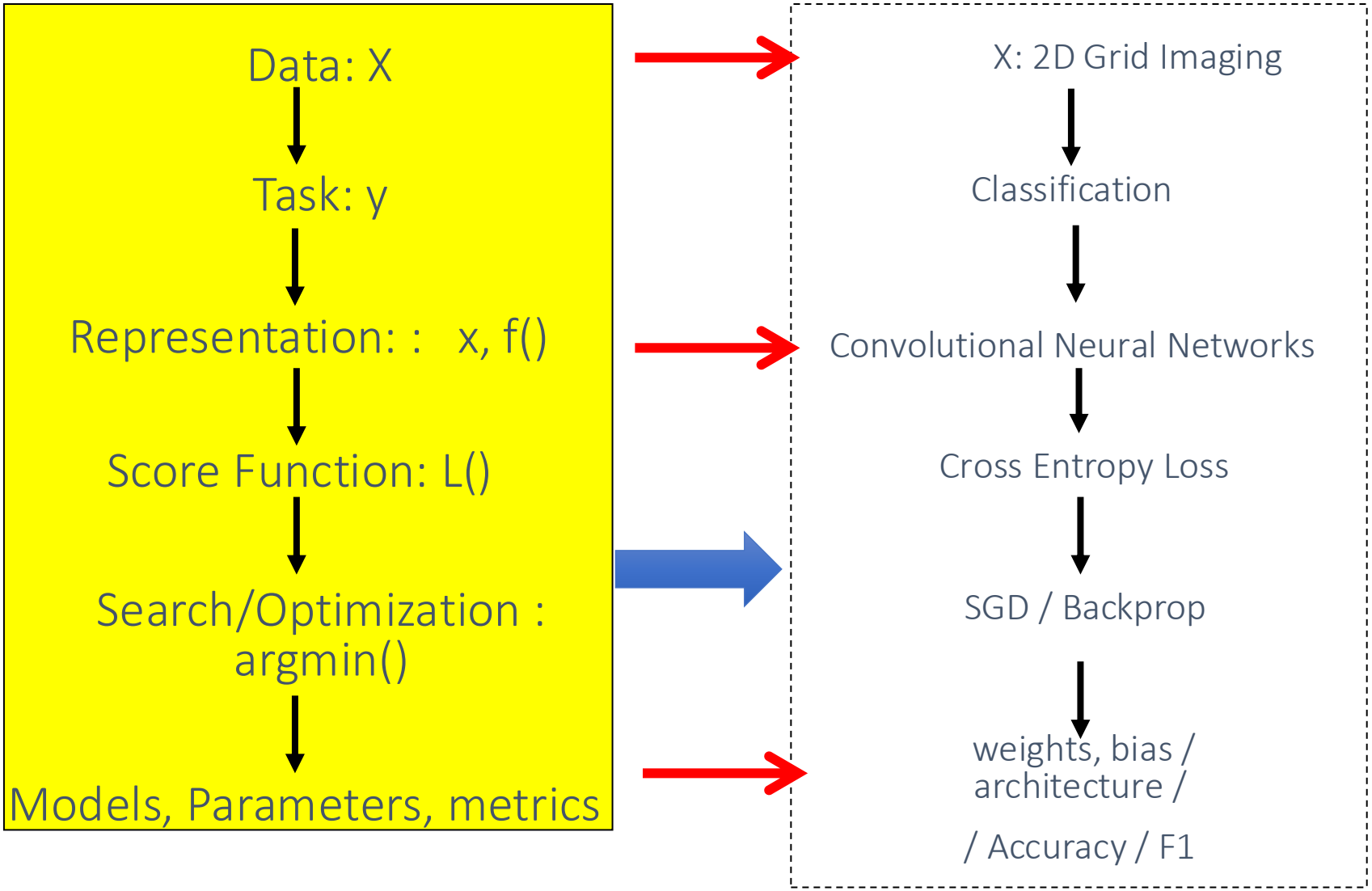


- 2010-11: hand-crafted computer vision pipelines
- 2012-2016: ConvNets
  - 2012: AlexNet
    - major deep learning success
  - 2013: ZFNet
    - improvements over AlexNet
  - 2014
    - VGGNet: deeper, simpler
    - InceptionNet: deeper, faster
  - 2015
    - ResNet: even deeper
  - 2016
    - ensembled networks
  - 2017
    - Squeeze and Excitation Network

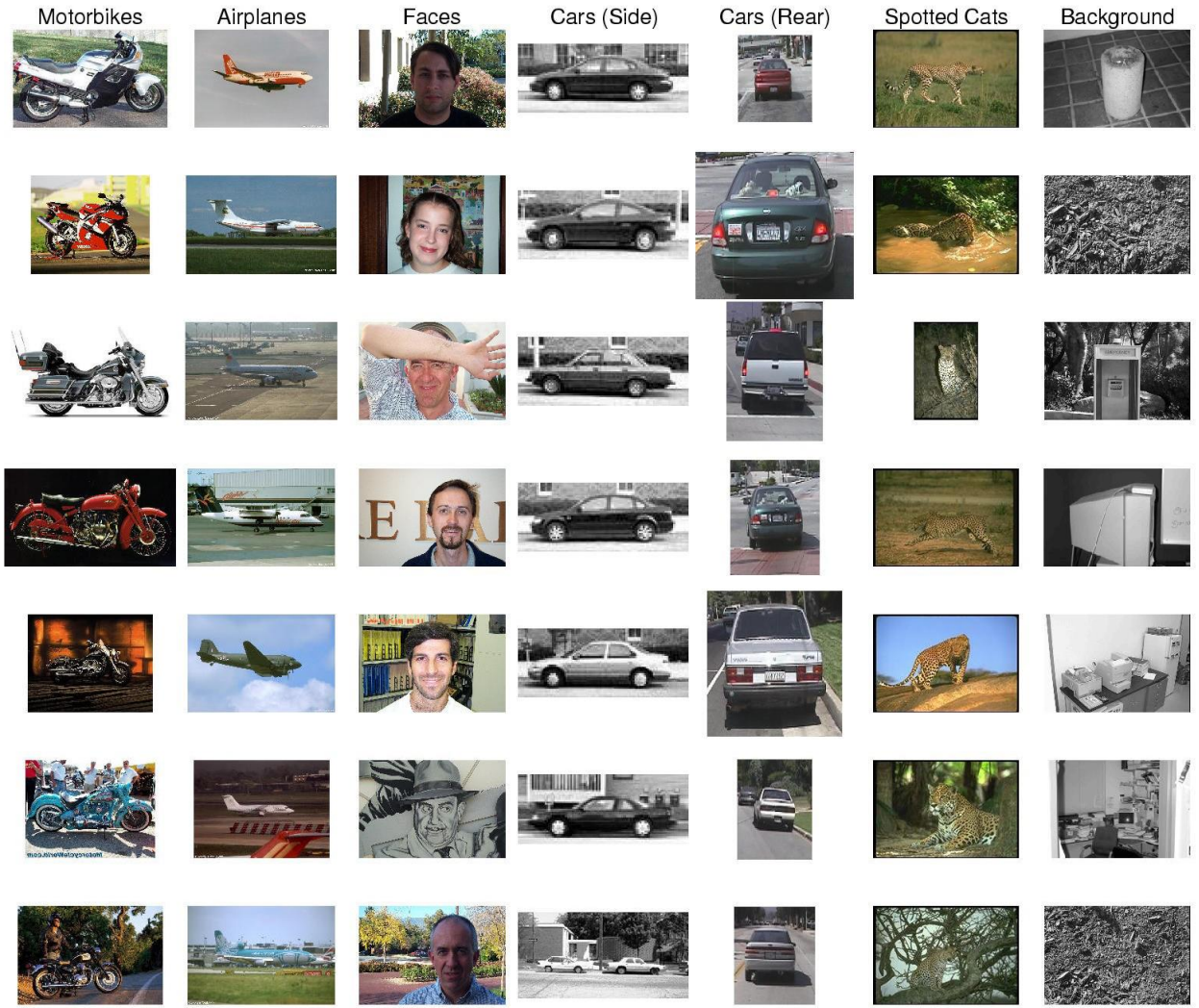


Adapt from From NIPS 2017 DL Trend Tutorial

Today: Convolutional Network Models on 2D Grid / Image



Focus: 2D Images for Classification

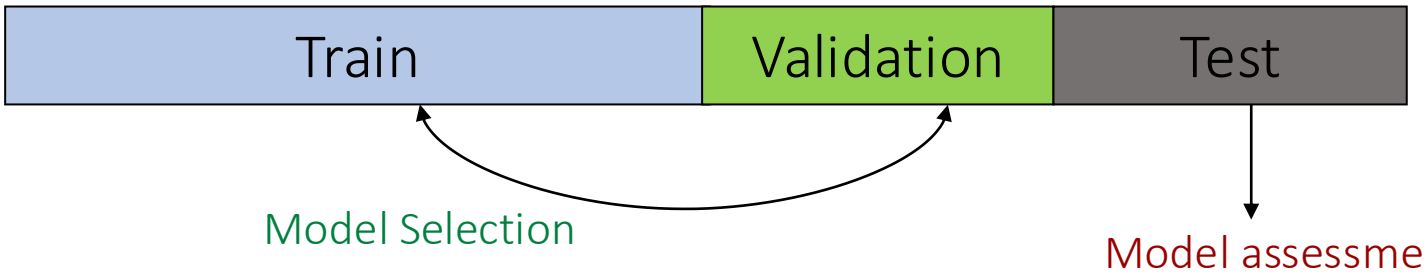


Review: Model Selection and Assessment

- Model Selection
  - Estimating performances of different models to choose the best one using validation data

- Model Assessment
  - Having chosen a model, estimating the prediction error on new data

testing



Review: Model Selection (Hyperparameter Tuning) & Model Assessment Pipelines in HW2

- (1) train / Validation / test →
- (2) k-CV on train to choose hyperparameter / then test → more expensive

		actual +	actual −
	predicted+	*TP*	*FP*
	predicted−	*FN*	*TN*

Binary Class {T, F}

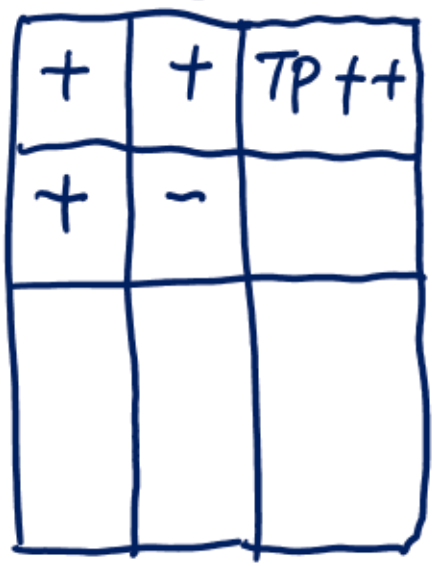
$\hat{y}$ vs y

- (number of) true positive (TP)
- (number of) true negative (TN)
- (number of) false positive (FP)
- (number of) false negative (FN)

	actual y	
	+	−
predicted+ ($\hat{y}$)	*TP*	*FP*
predicted−	*FN*	*TN*

Confusion matrix

- (number of) true positive (TP)
- (number of) true negative (TN)
- (number of) false positive (FP)
- (number of) false negative (FN)



$\hat{y}_1$ vs y_1

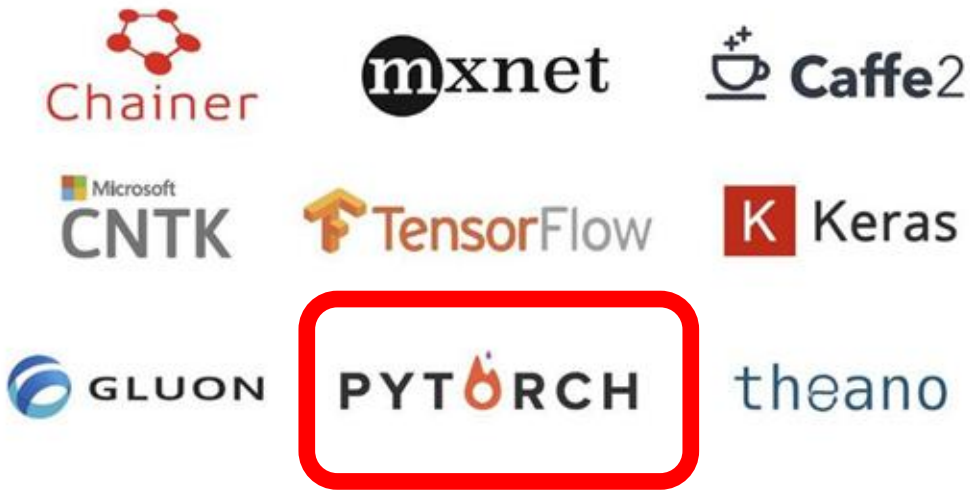
$\hat{y}_2$ y_2

$\hat{y}_m$ vs y_n

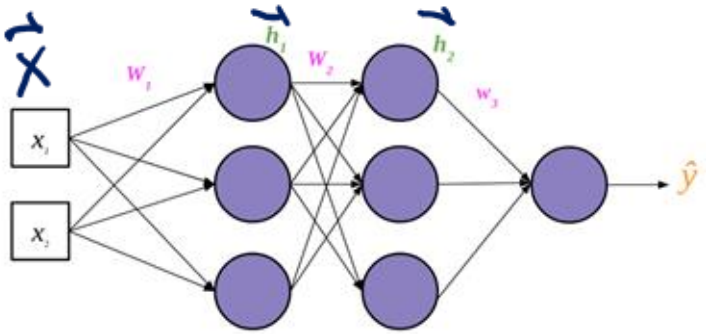
Metric	Formula	Interpretation
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Overall performance of model
Precision	$\frac{TP}{TP + FP}$	How accurate the positive predictions are
Recall Sensitivity	$\frac{TP}{TP + FN}$	Coverage of actual positive sample
Specificity	$\frac{TN}{TN + FP}$	Coverage of actual negative sample
F1 score	$\frac{2TP}{2TP + FP + FN}$	Hybrid metric useful for unbalanced classes

Credit: Stanford Machine Learning course

Deep Learning Frameworks



Pytorch Sample Code




```python
import torch.nn as nn
import torch.nn.functional as F

class ThreeLayerNet(torch.nn.Module):
    def __init__(self, d_in, d_hidden, d_out):
        super().__init__()
        self.W1 = nn.Linear(d_in,d_hidden)
        self.W2 = nn.Linear(d_hidden,d_hidden)
        self.w3 = nn.Linear(d_hidden,d_out)
        self.nonlinear = nn.Sigmoid()

    def forward(self, x):
        h1 = self.nonlinear(self.W1(x))
        h2 = self.nonlinear(self.W2(h1))
        y_hat = self.nonlinear(self.w3(h2))
        return y_hat

model = ThreeLayerNet(2,3,1)
```

UVA CS 4774: Machine Learning

Lecture 13: Supervised Image Classification and Convolutional Neural Networks

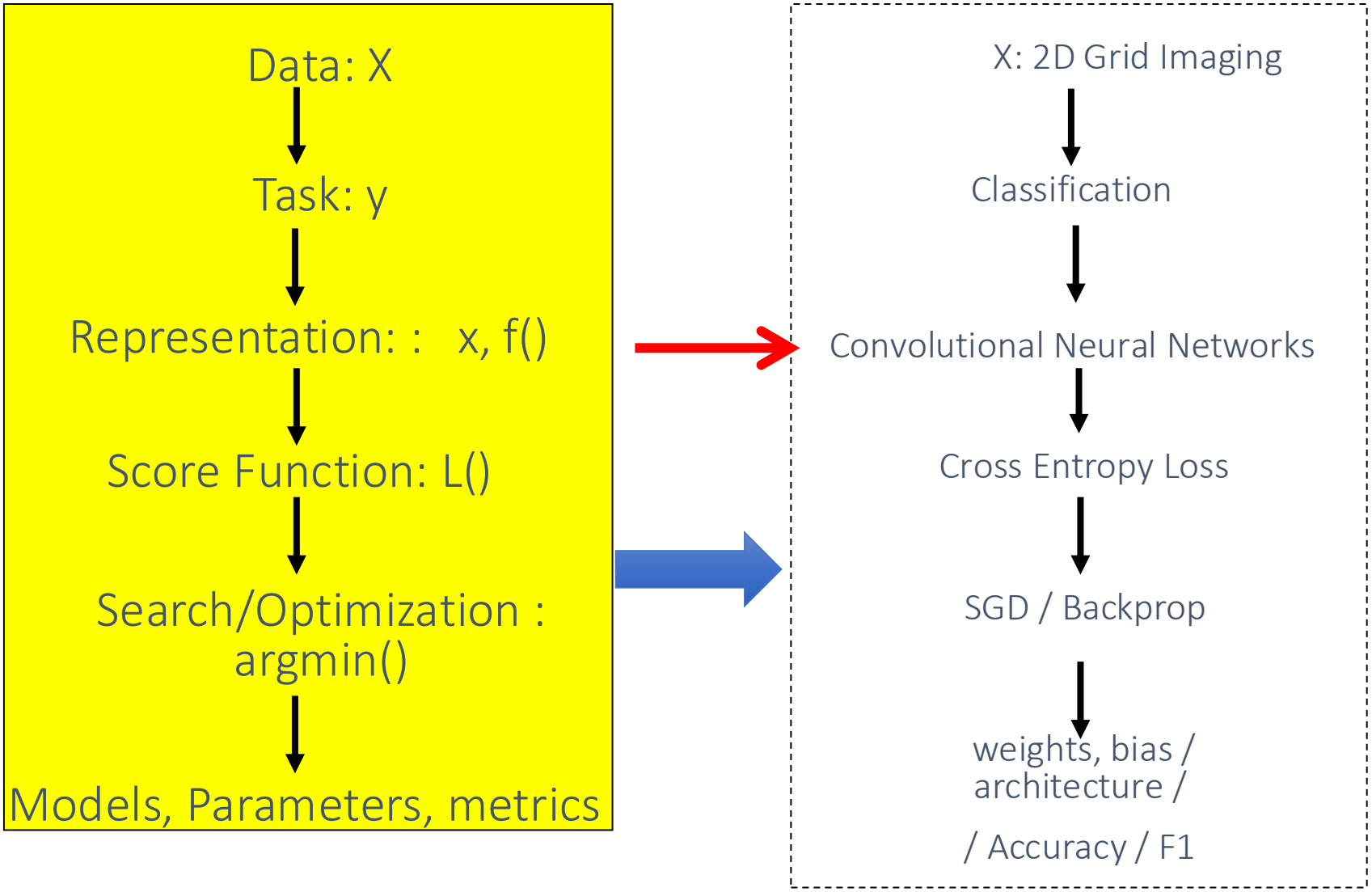
Dr. Yanjun Qi

University of Virginia

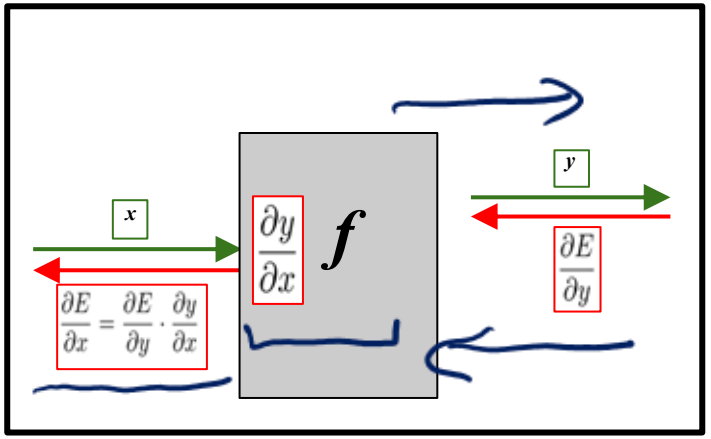
Department of Computer Science

Module II

Today: Convolutional Network Models on 2D Grid / Image



Building Deep Neural Nets







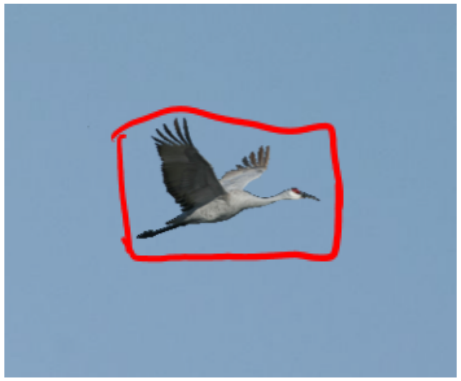


http://cs231n.stanford.edu/slides/winter1516_lecture5.pdf

Important Block: Convolutional Neural Networks (CNN)

- Prof. Yann LeCun invented CNN in 1998
- First NN successfully trained with many layers







The bird occupies a local area and looks the same in different parts of an image.
We should construct neural nets which exploit these properties!

Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, Gradient-based learning applied to document recognition, Proceedings of the IEEE 86(11): 2278–2324, 1998.

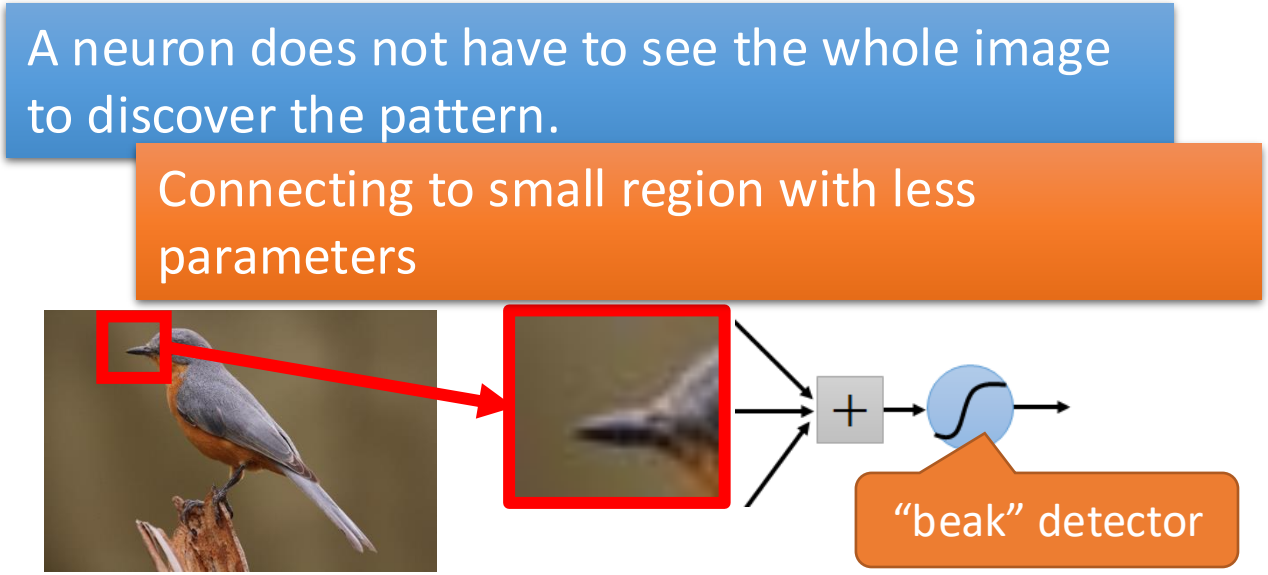
Adapt from From NIPS 2017 DL Trend Tutorial

Locality and Translation Invariance

- **Locality**: objects tend to have a local spatial support
- **Translation invariance**: object appearance is independent of location
- Can define these properties since an image lies on a grid/lattice
  - ConvNet applicable to other data with such properties, e.g. audio/text
  - Lattice: regular spacing or arrangement of geometric points;

Property of Object Recognition in Images

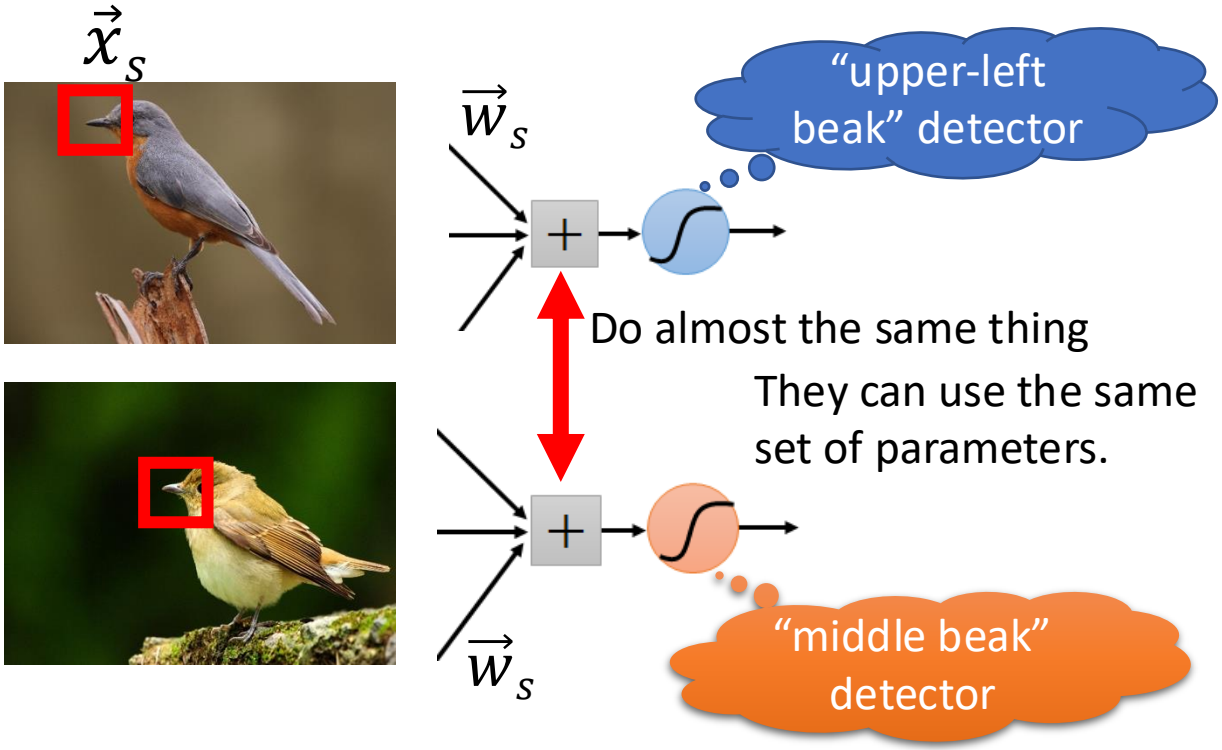
- (1) Locality: Some patterns are much smaller than the whole image



Dr. Hung-yi Lee’s CNN slides

Property of Object Recognition in Images

- (2) Translation invariance: The same patterns appear in different regions.



Dr. Hung-yi Lee's CNN slides

Property of Object Recognition in Images

- (3) Subsampling the pixels will not change the object

bird



subsampling

bird



We can subsample the pixels to make image smaller

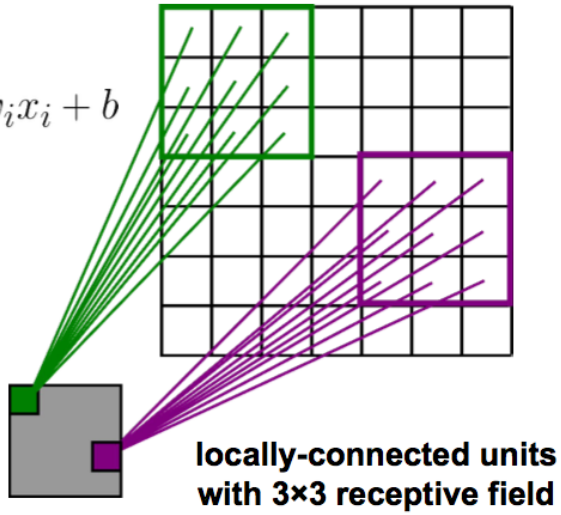
→ Less parameters for the network to process the image

Dr. Hung-yi Lee's CNN slides

Summary: CNN models Locality and Translation Invariance

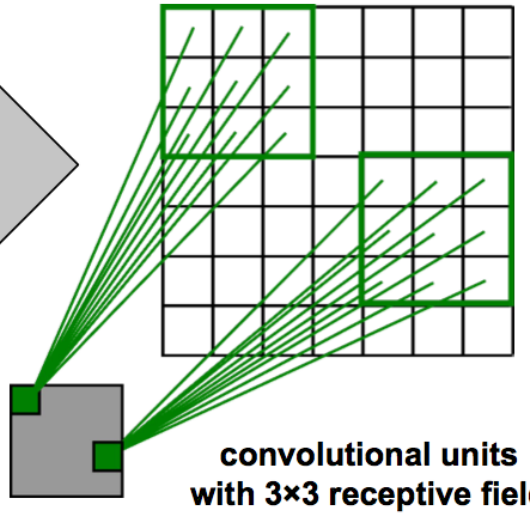
Make fully-connected layer locally-connected and sharing weight

$$y = \sum_{i \in receptive\ field} w_i x_i + b$$



locally-connected units
with 3×3 receptive field

weight sharing



convolutional units
with 3×3 receptive field

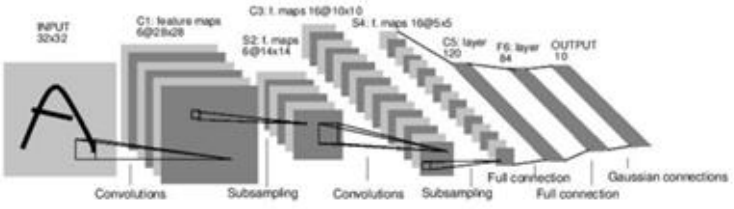
$$y = w * x + b$$

Adapt from From NIPS 2017 DL Trend Tutorial

History of ConvNets

1998

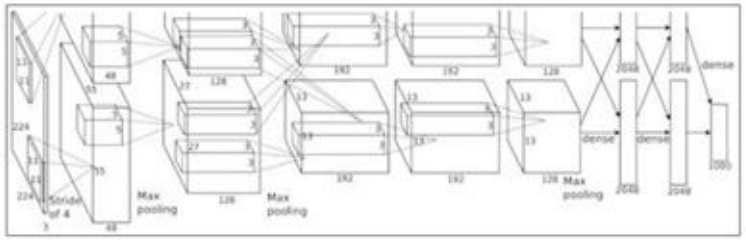
Gradient-based learning applied to document recognition [LeCun, Bottou, Bengio, Haffner]



LeNet-5

2012

ImageNet Classification with Deep Convolutional Neural Networks [Krizhevsky, Sutskever, Hinton, 2012]

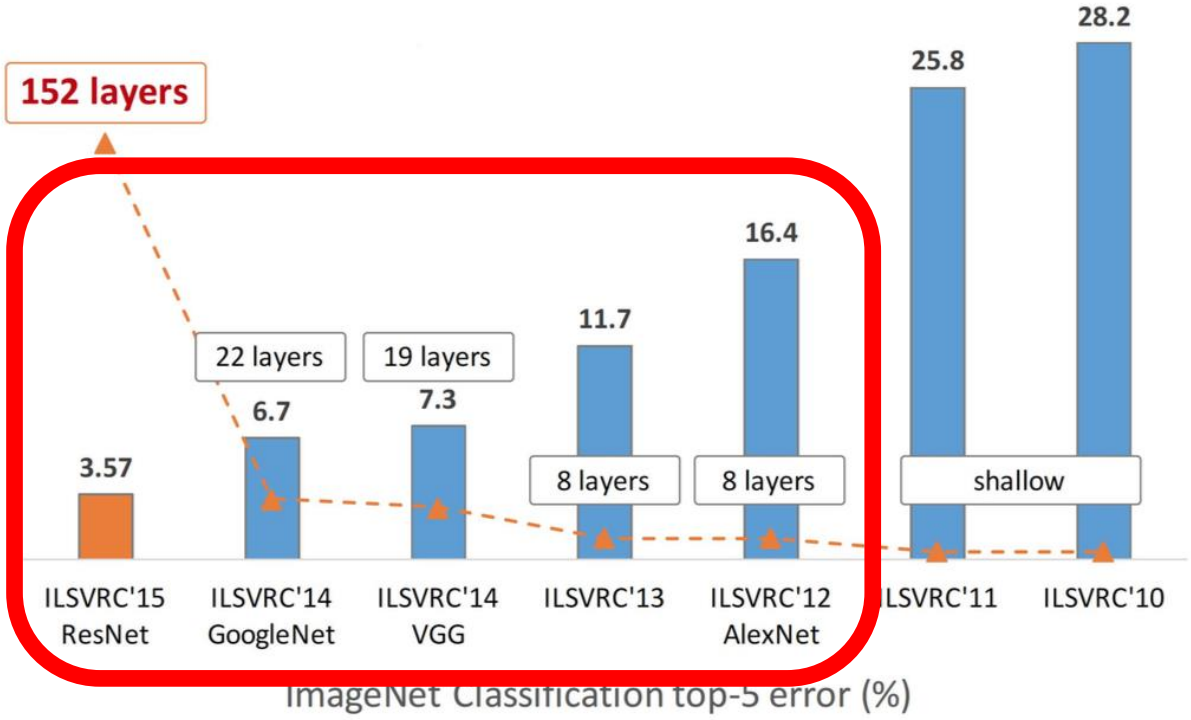


"AlexNet"

Revolution of Depth

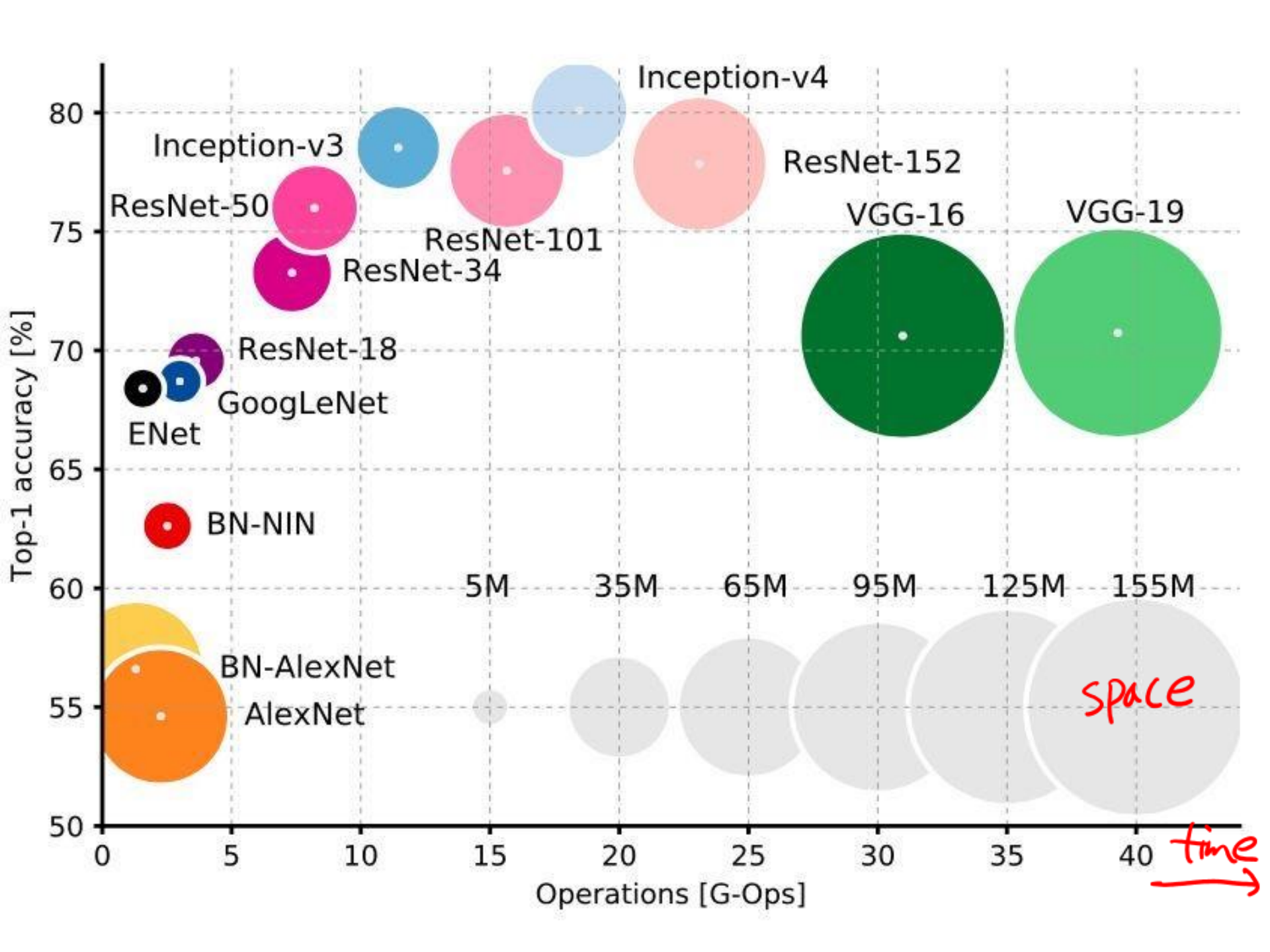




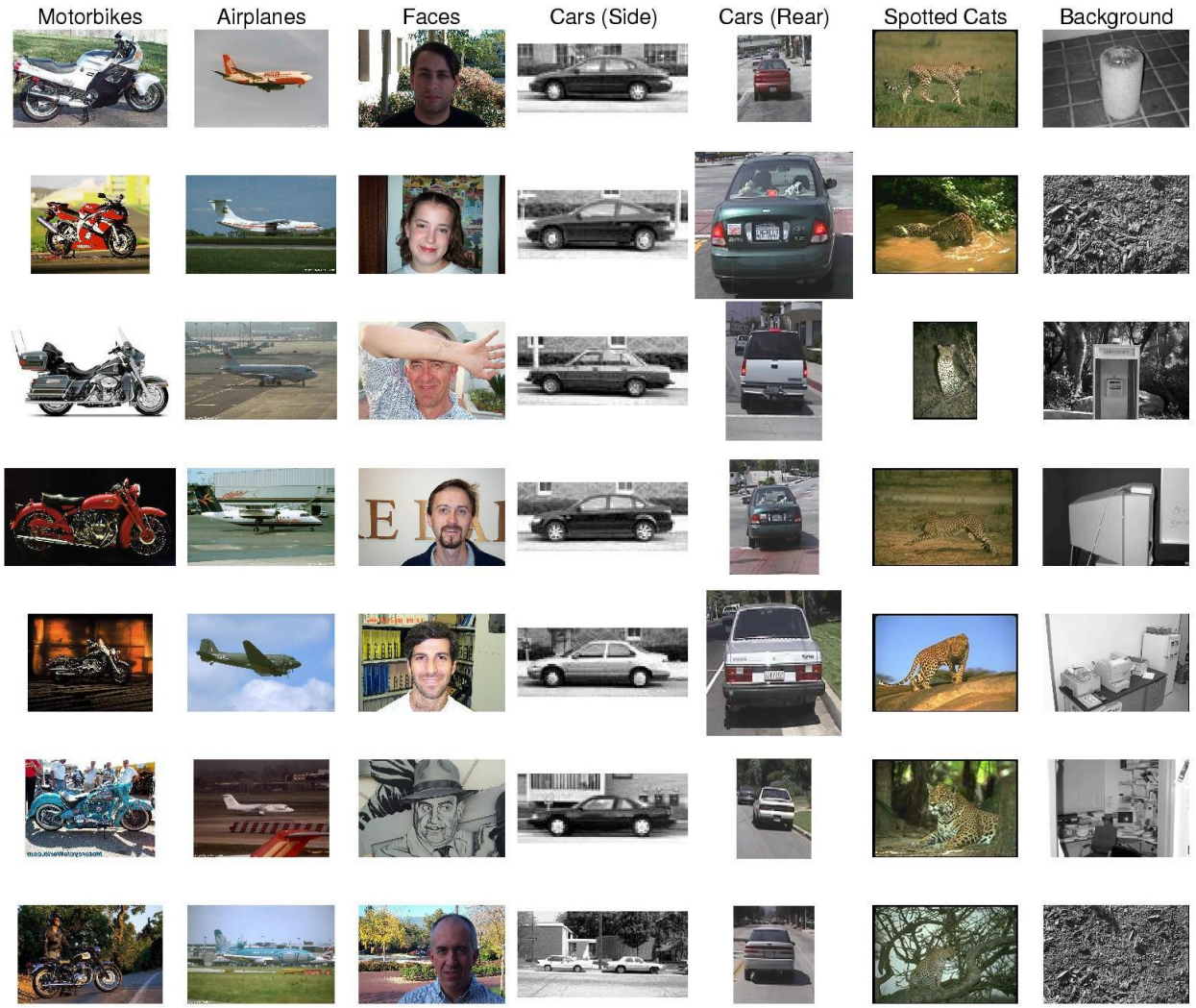


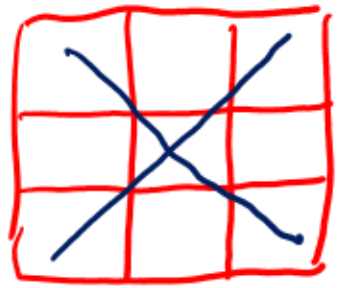
Kaiming He, Xiangyu Zhang, Shaoqing Ren, & Jian Sun. "Deep Residual Learning for Image Recognition". CVPR 2016.

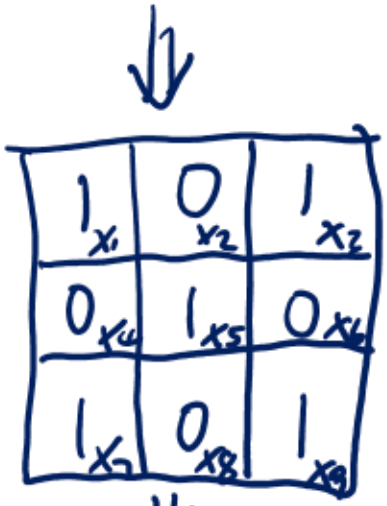
Adapt from From NIPS 2017 DL Trend Tutorial

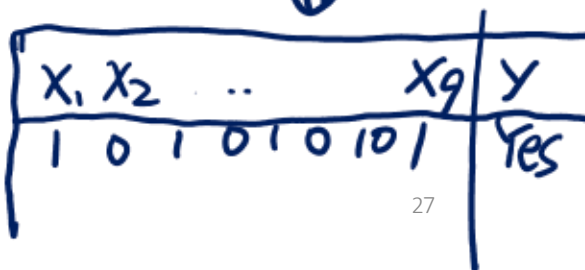

Inception-v4
Inception-v3
ResNet-152
ResNet-50
VGG-16
VGG-19
ResNet-101
ResNet-34
ResNet-18
GoogLeNet
ENet
BN-NIN
5M
35M
65M
95M
125M
155M
BN-AlexNet
AlexNet
space
time
Top-1 accuracy [%]
Operations [G-Ops]
80
75
70
65
60
55
50
0
5
10
15
20
25
30
35
40

Focus: 2D Images for Classification









CNN – Convolution

The following two are the network parameters to be learned.

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

"detector 1"

Filter 1

Matrix $\vec{w}_{s1}$

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

Matrix $\vec{w}_{s2}$

"detector 2"

Property 1 Each filter detects a small pattern (3 x 3).

CNN – Convolution

Those are the network parameters to be learned.

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

Matrix $\vec{w}_{s1}$

"detector 1"

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

Matrix $\vec{w}_{s2}$

"detector 2"

Property 1 Each filter detects a small pattern (3 x 3).

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CNN – Convolution

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

3 -1

CNN – Convolution

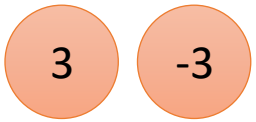
1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

If stride=2

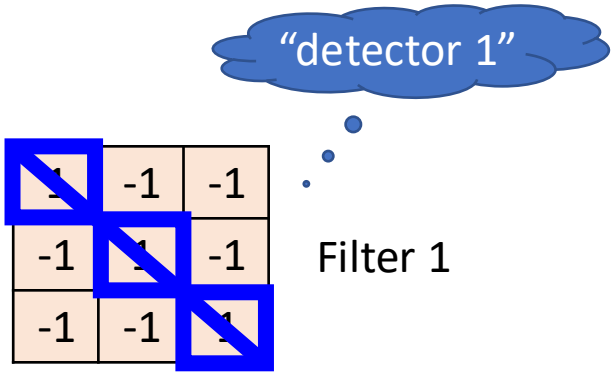
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

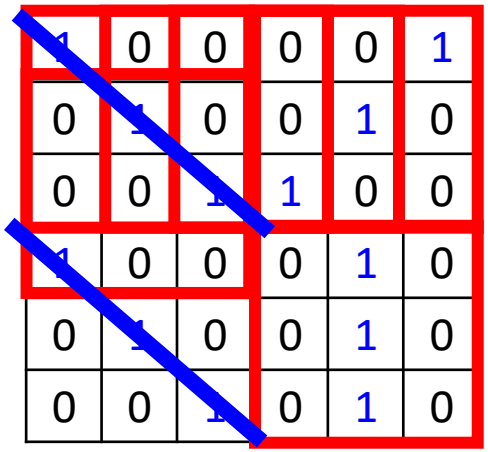


We set stride=1 below

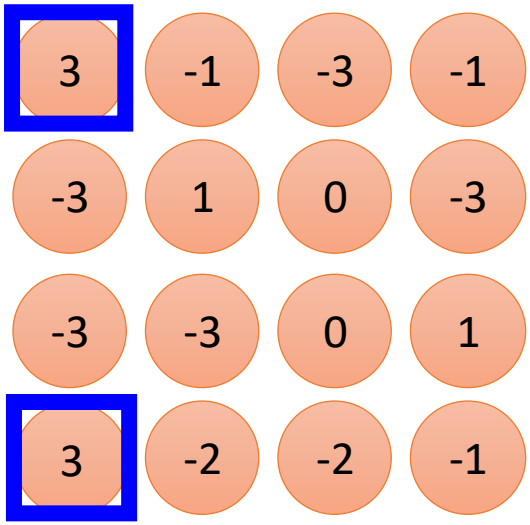
CNN – Convolution



stride=1



6 x 6 image

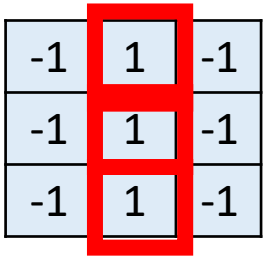


Property 2

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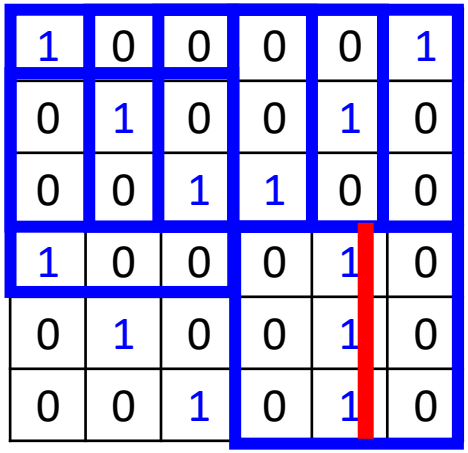
CNN – Convolution





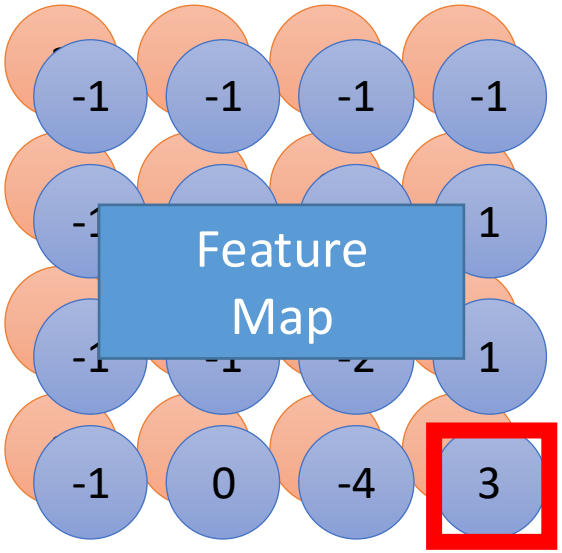
Filter 2

stride=1



6 x 6 image

Do the same process for every filter



4 x 4 image

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CNN – Convolution

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

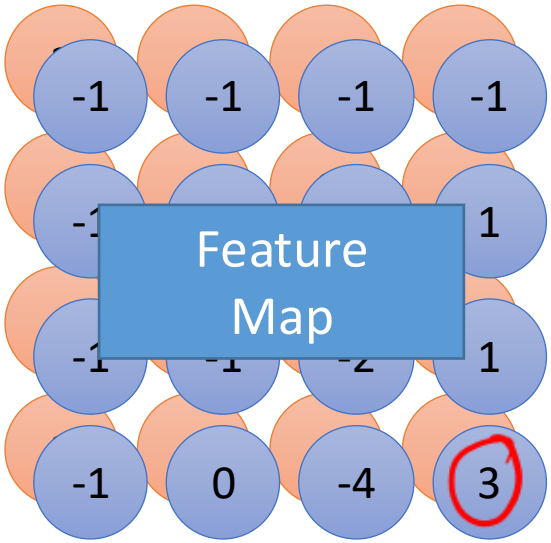
Filter 2

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

You can do the same process for every filter

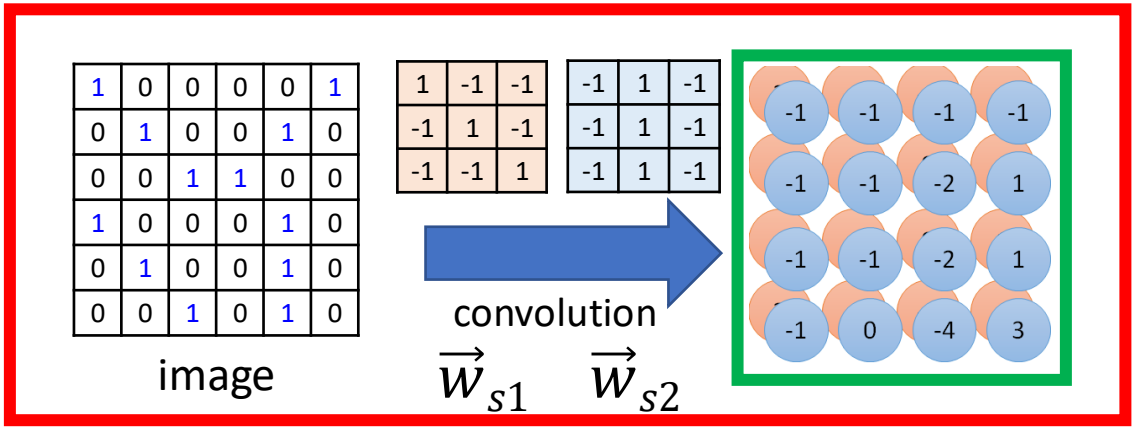


4 x 4 image

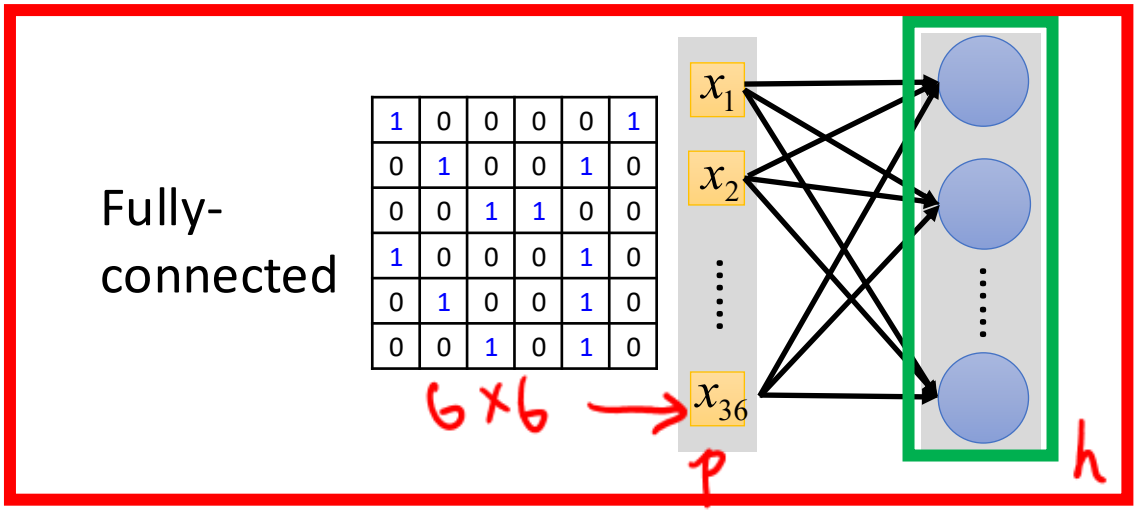
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Convolution v.s. Fully Connected

Convolution



Fully Connected

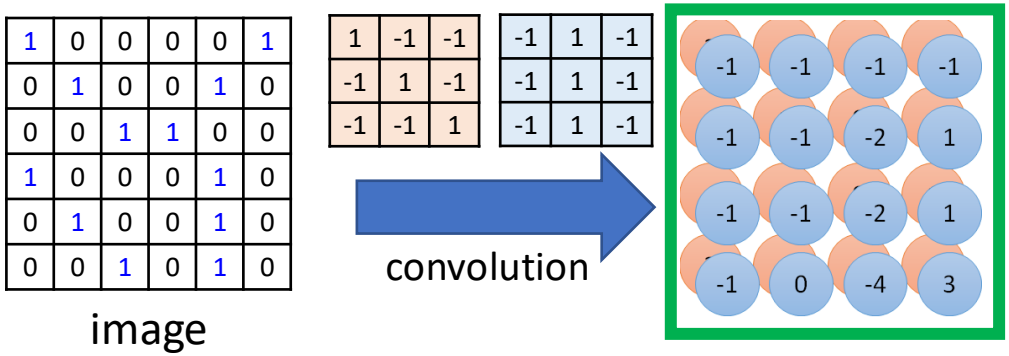


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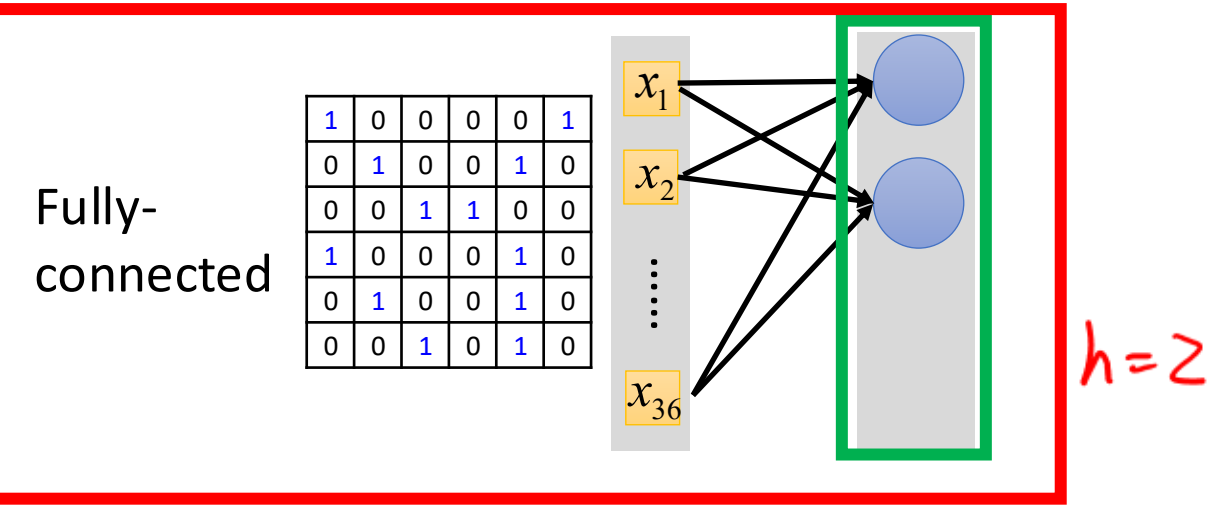
Convolution v.s. Fully Connected

When with 2 filters, 3*3*2=18 parameters!

Convolution



Fully Connected

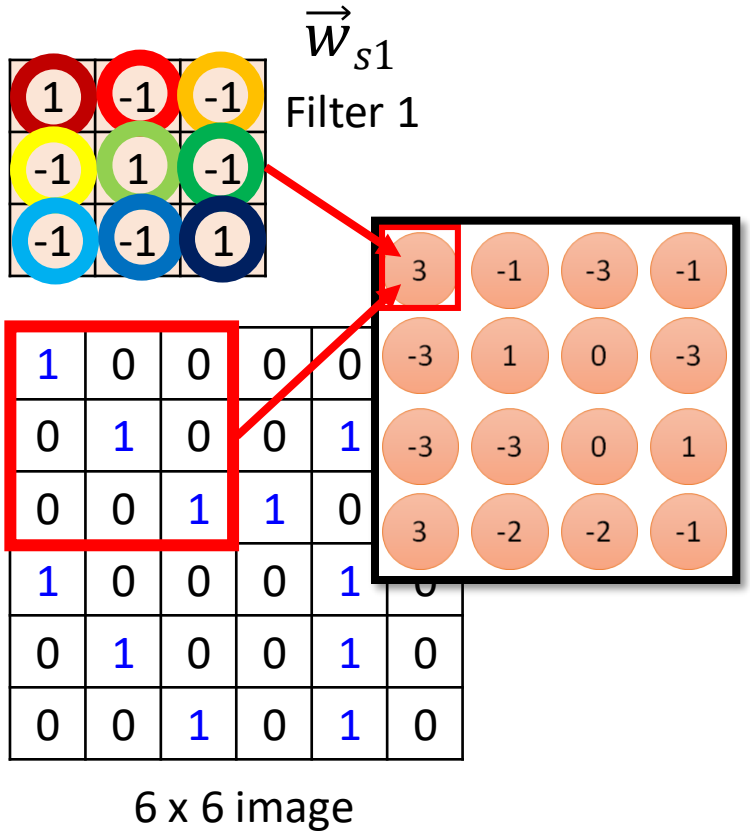


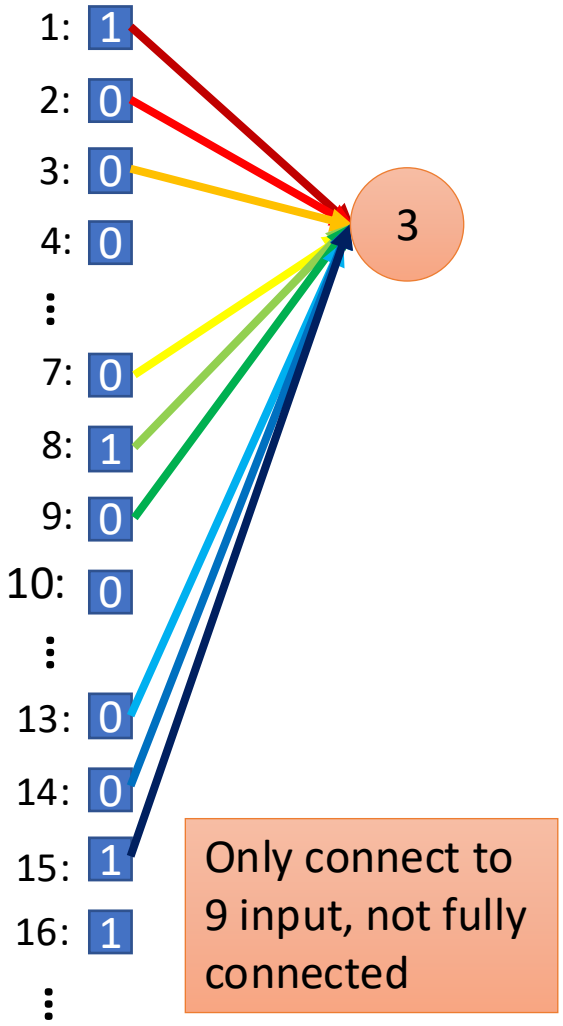
When 2 filters, 36*2=72 parameters!

(1) Locality:

Less parameters!

Each filter has 3*3=9 parameters!



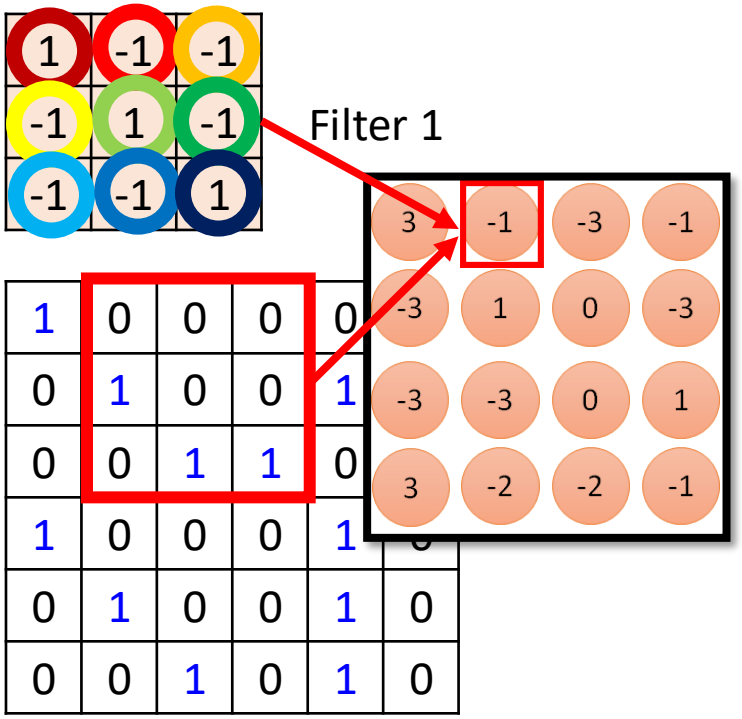


Only connect to 9 input, not fully connected

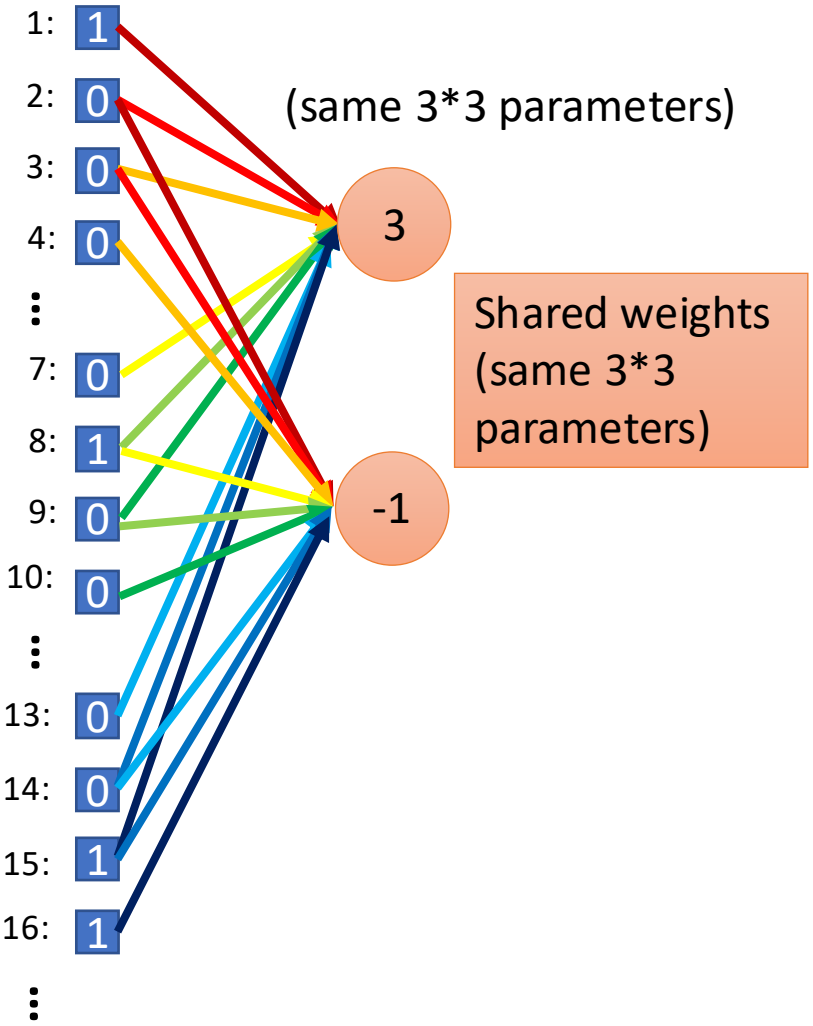
Locality: Some patterns are much smaller than the whole image

Dr. Hung-yi Lee's CNN slides

(2) Translation invariance:



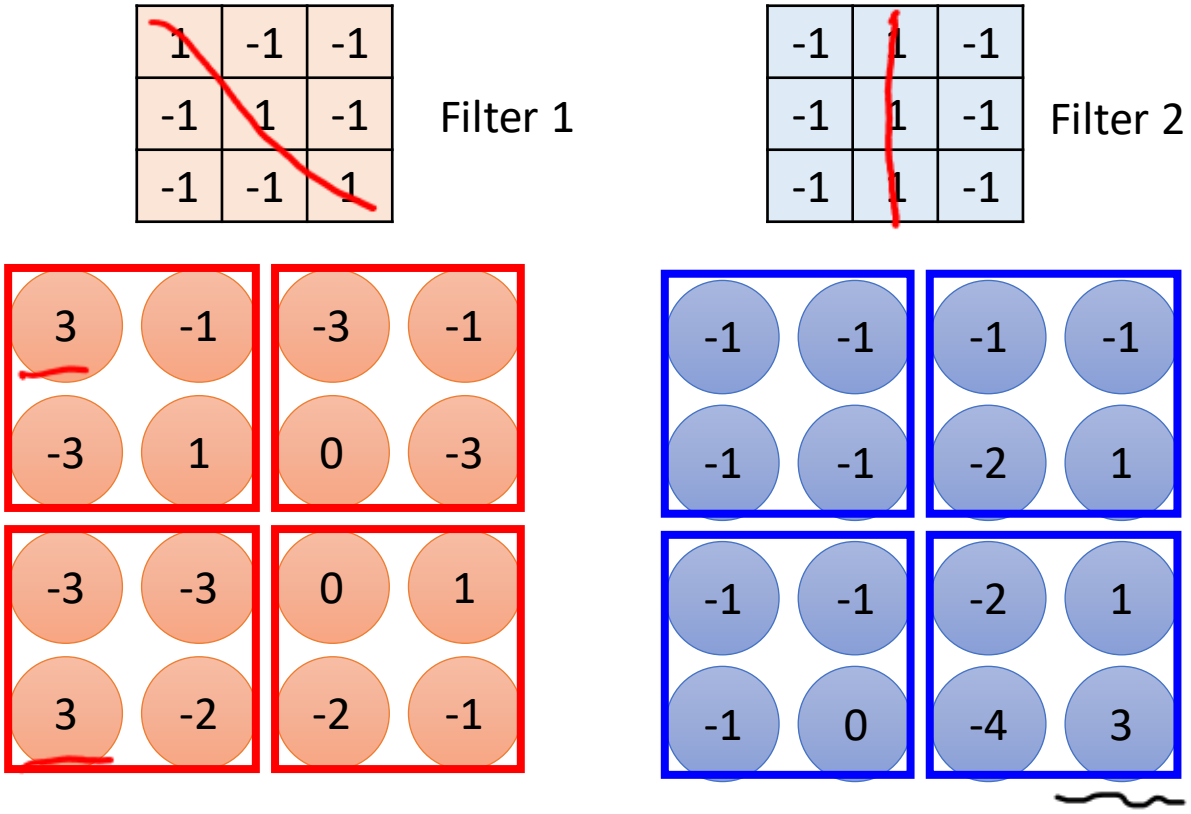
Less parameters!



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(3) Subsampling:

CNN – Max Pooling



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(3) Subsampling:

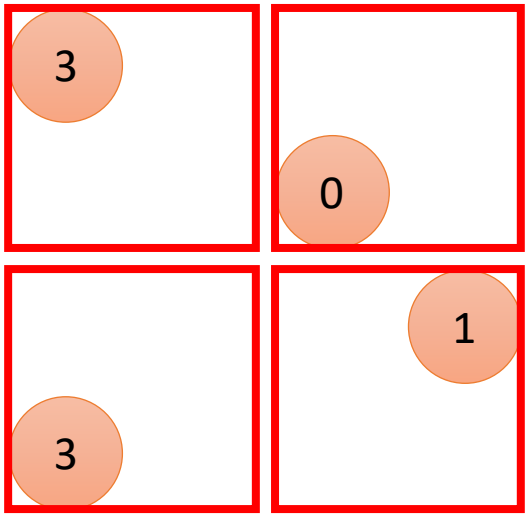
CNN – Max Pooling

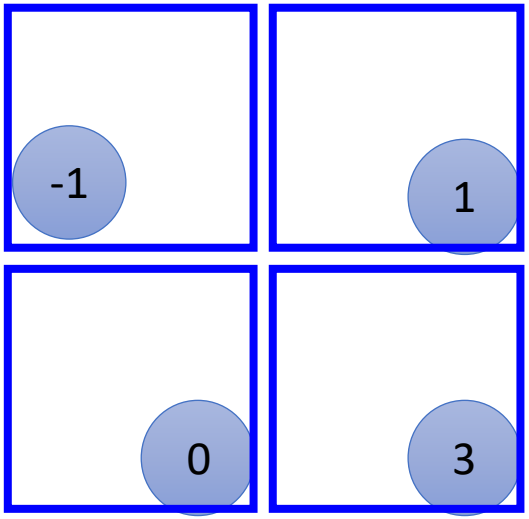
1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

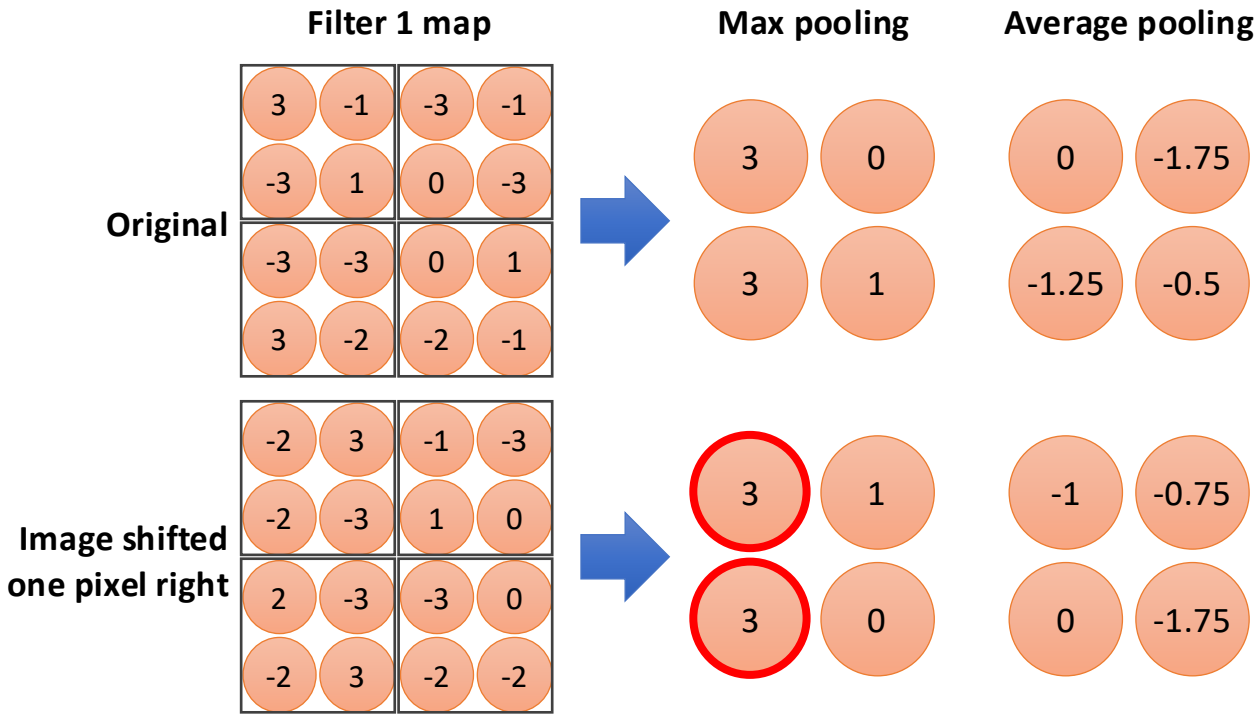




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(3) Subsampling:

Max vs. Average Pooling



Top-left group:
max keeps the 3,
the mean is 0

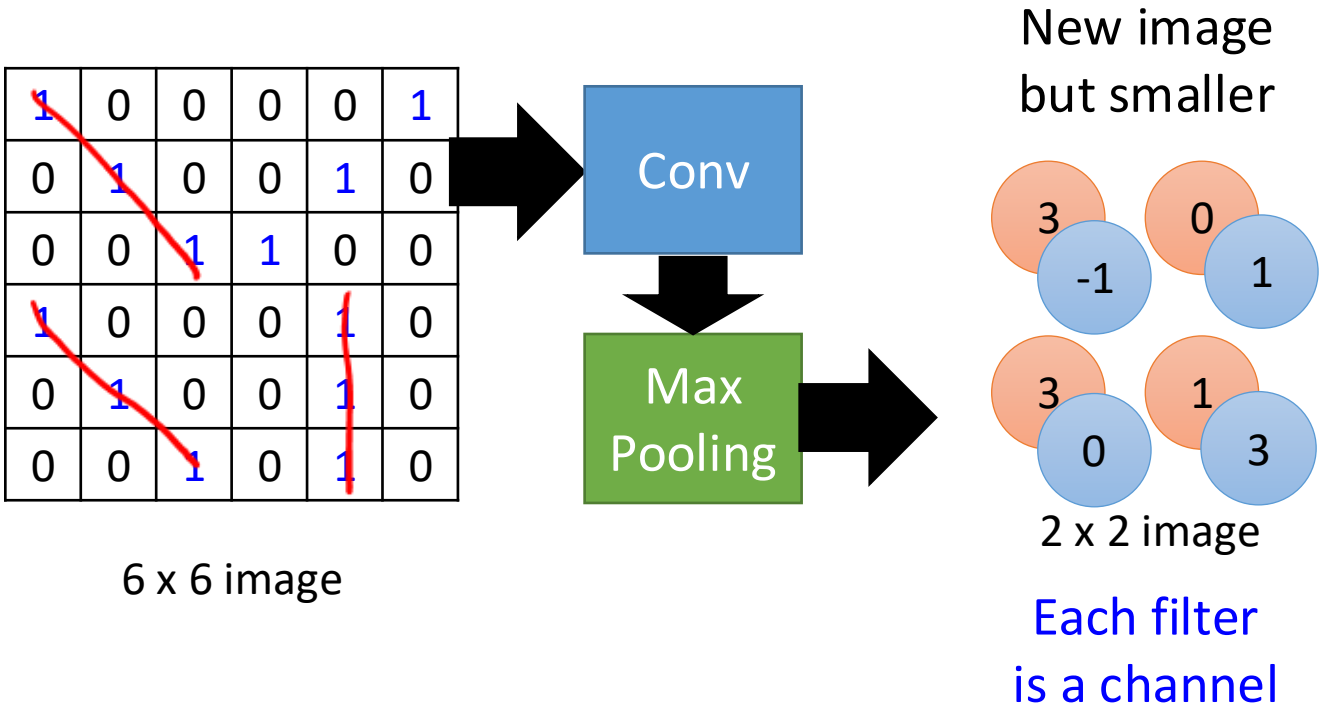
Red = unchanged
Max: 2 of 4 stay the same
Average: 0 of 4

Max reports the strongest response in each group, so a strong match survives
Average reports the overall response, so values can cancel out
A shift that stays inside a group leaves max unchanged; a bigger one can change it

Map: Dr. Hung-yi Lee's CNN slides; layout after Harvard CS109B; shift idea after Srihari CSE676

(3) Subsampling:

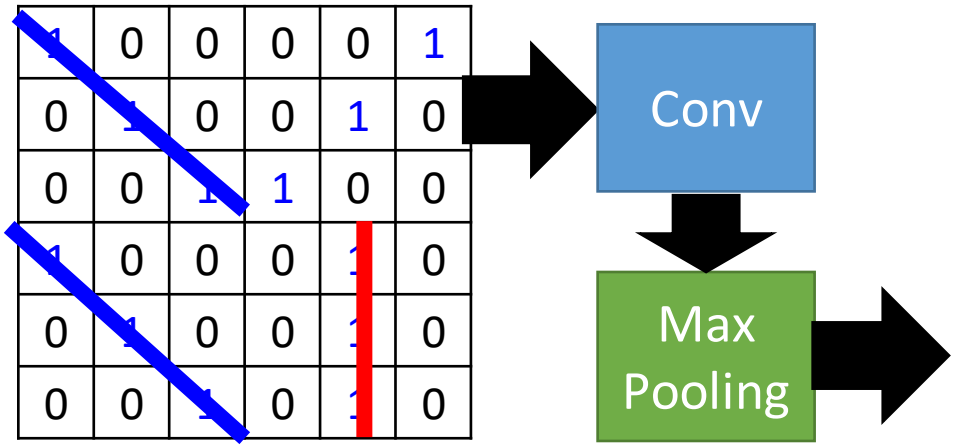
CNN – Max Pooling



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(3) Subsampling:

CNN – Max Pooling



6 x 6 image

New image but smaller

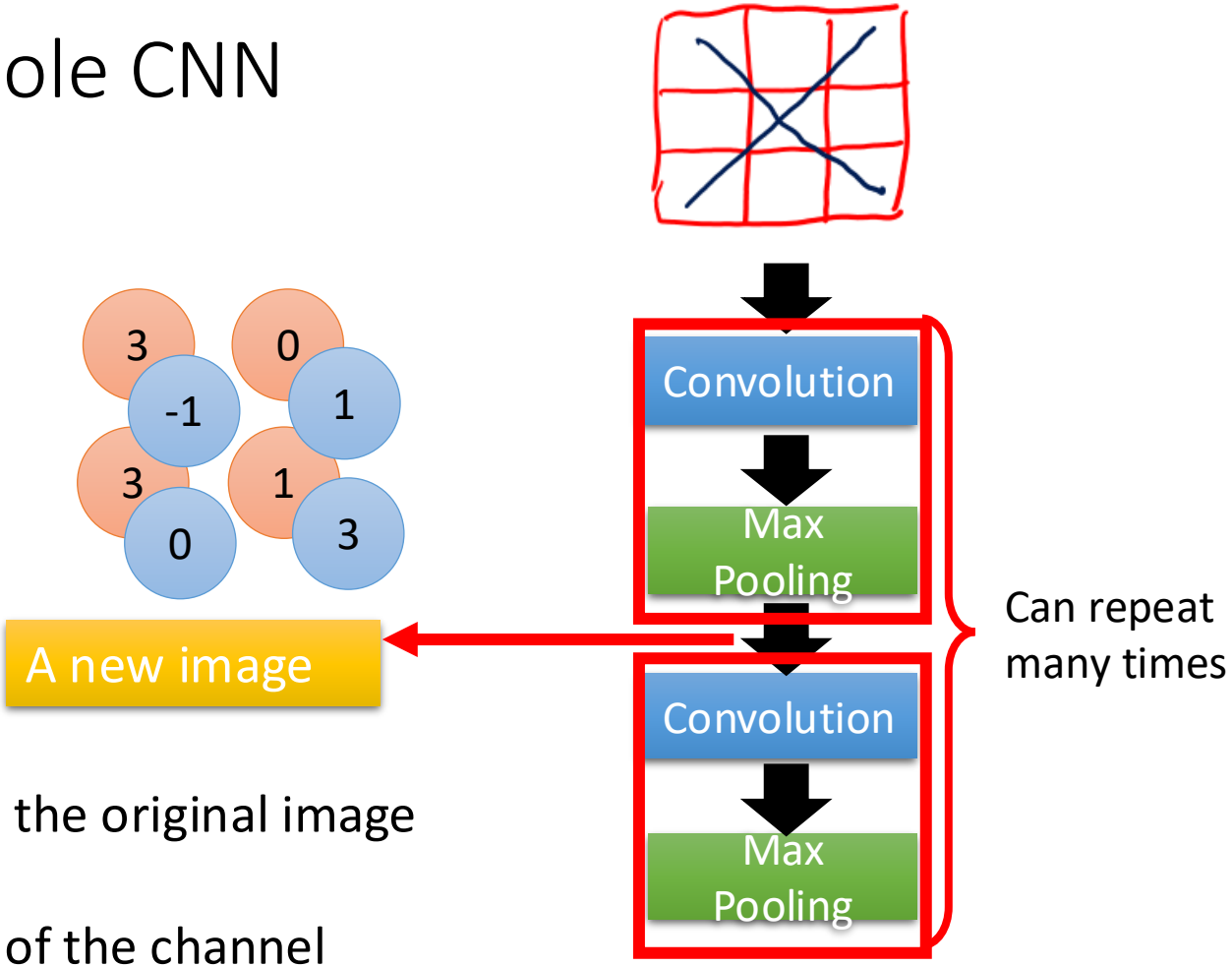
2 x 2 image

Each filter is a channel

6x6
3x3
4x4
2x2

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The whole CNN

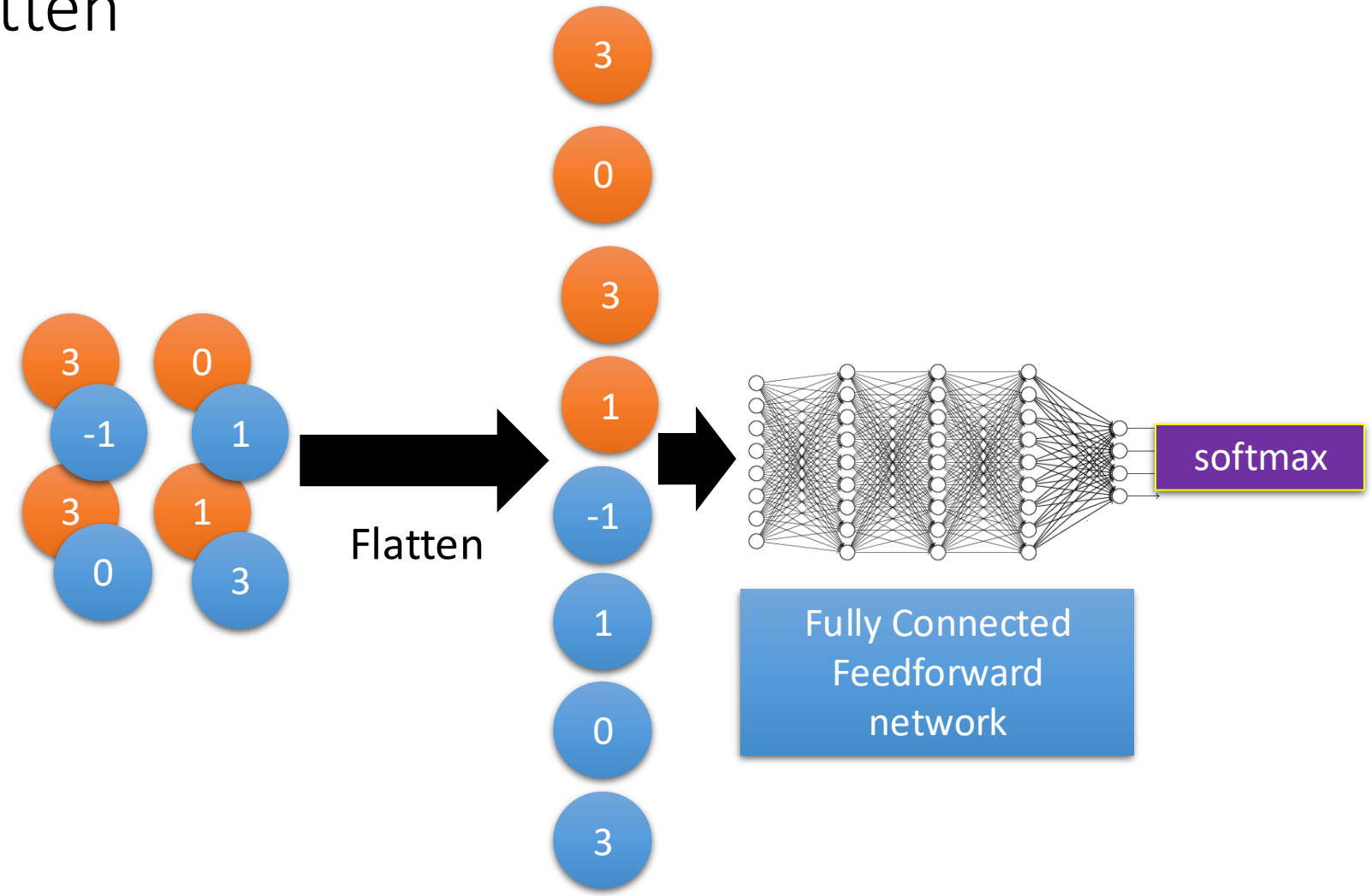


Smaller than the original image

The number of the channel is the number of filters

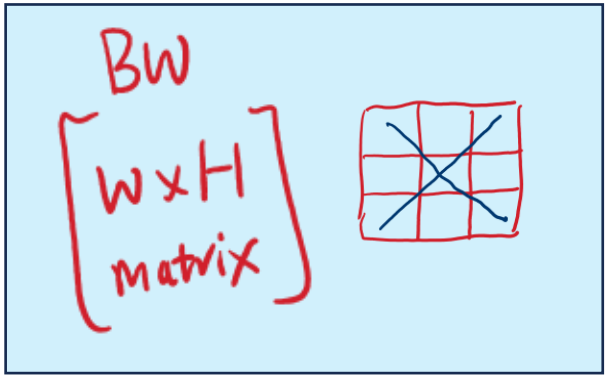
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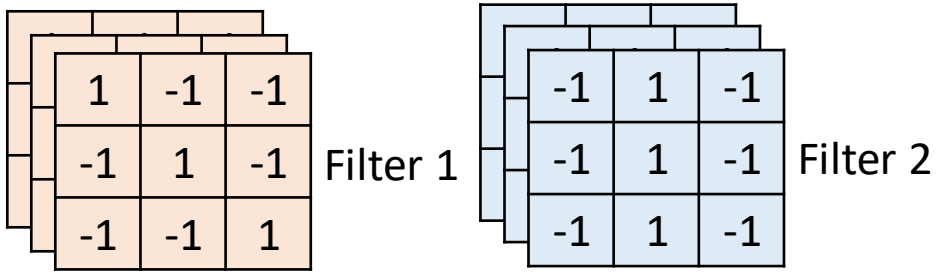
Flatten

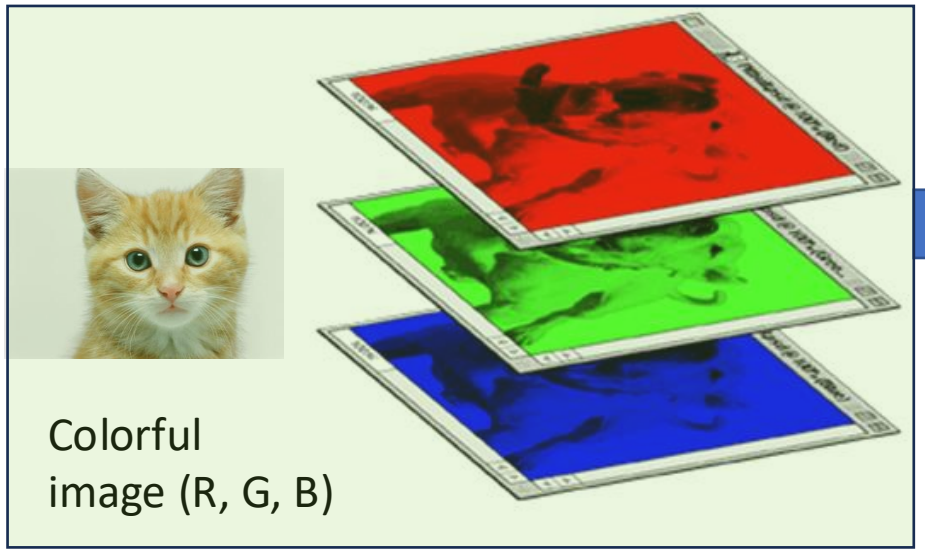


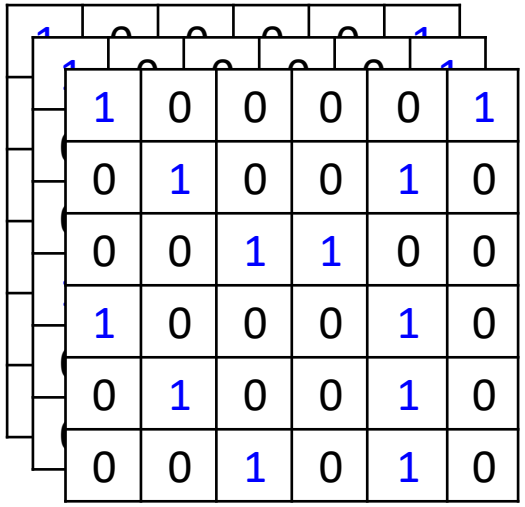
Dr. Hung-yi Lee's CNN slides

CNN – Colorful image (from matrix to tensor)



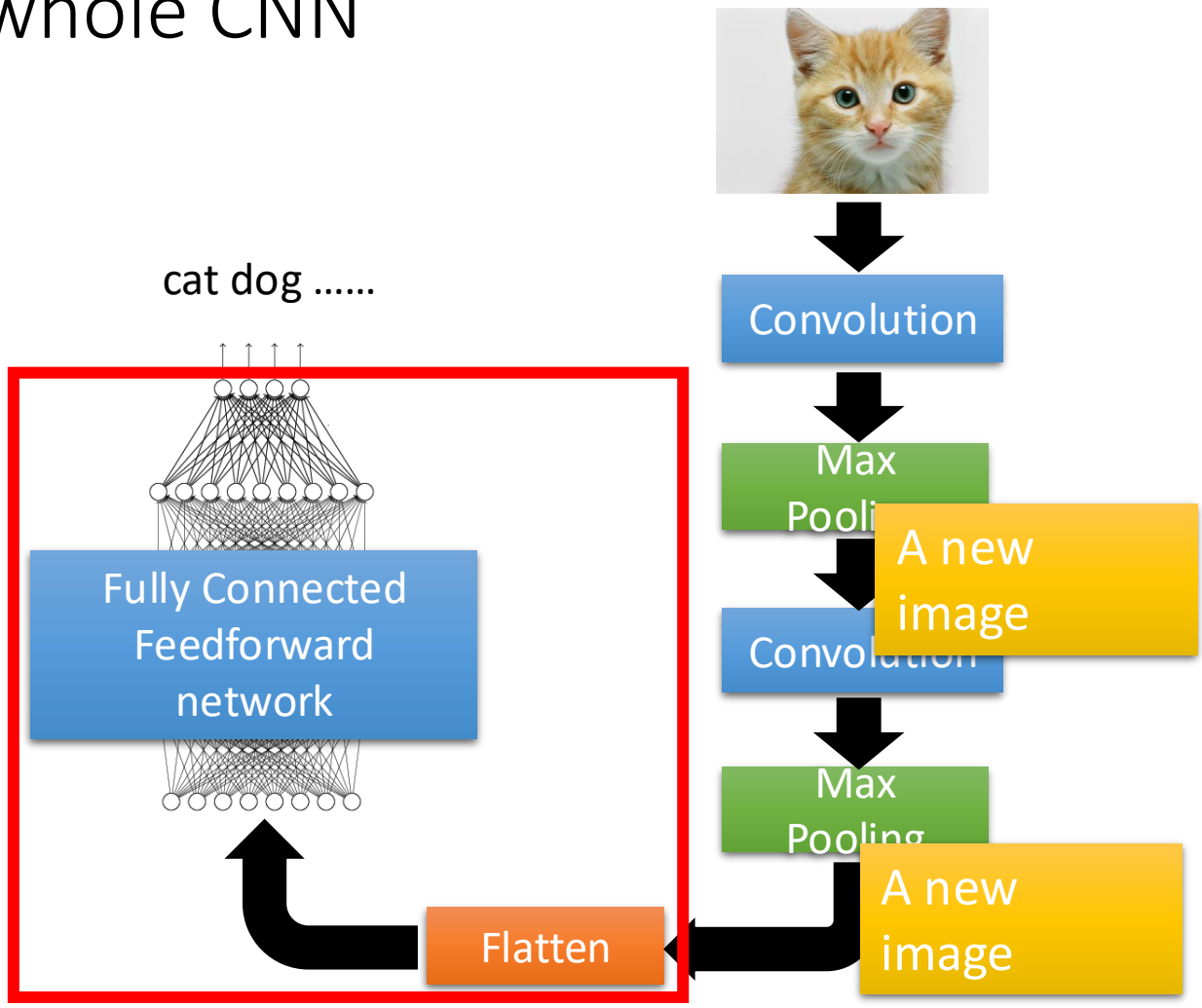






Dr. Hung-yi Lee's CNN slides

The whole CNN



Dr. Hung-yi Lee's CNN slides

How many filters?

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
0	0	0	1	0	0
0	0	0	1	0	0
0	0	0	1	0	0

6 x 6 image

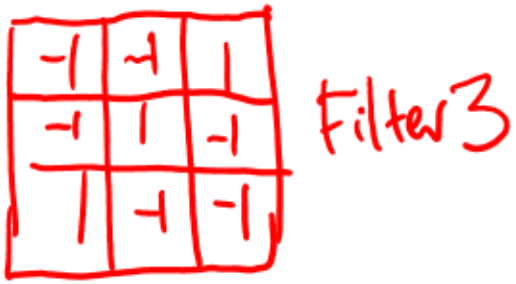
How many filters?

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

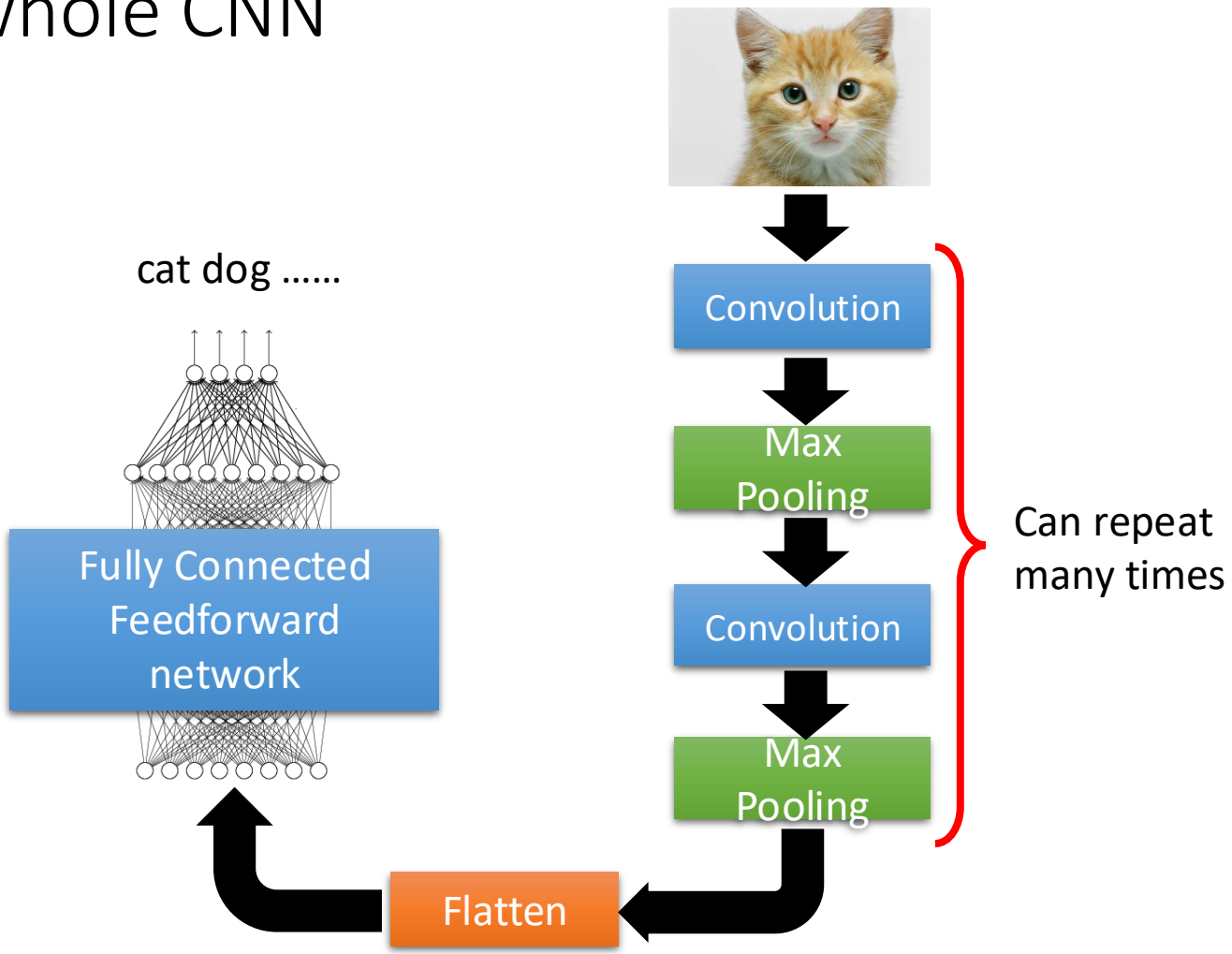


stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
0	0	0	1	0	0
0	0	0	1	0	0
0	0	0	1	0	0

6 x 6 image

The whole CNN



Dr. Hung-yi Lee's CNN slides

Summary: whole CNN



Property 1

- Some patterns are much smaller than the whole image

Property 2

- The same patterns appear in different regions.

Property 3

- Subsampling the pixels will not change the object

Convolution

Max Pooling

Convolution

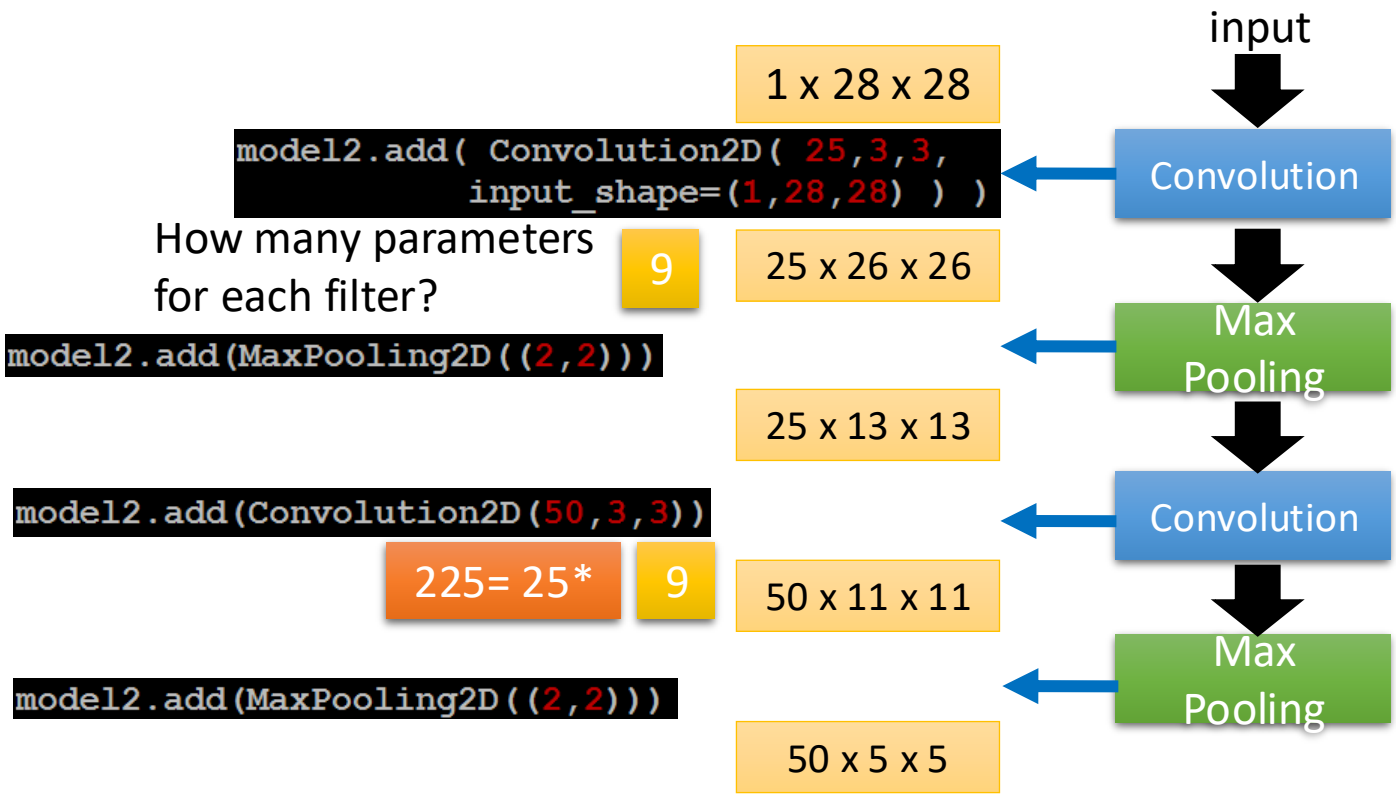
Max Pooling

Can repeat many times

Flatten

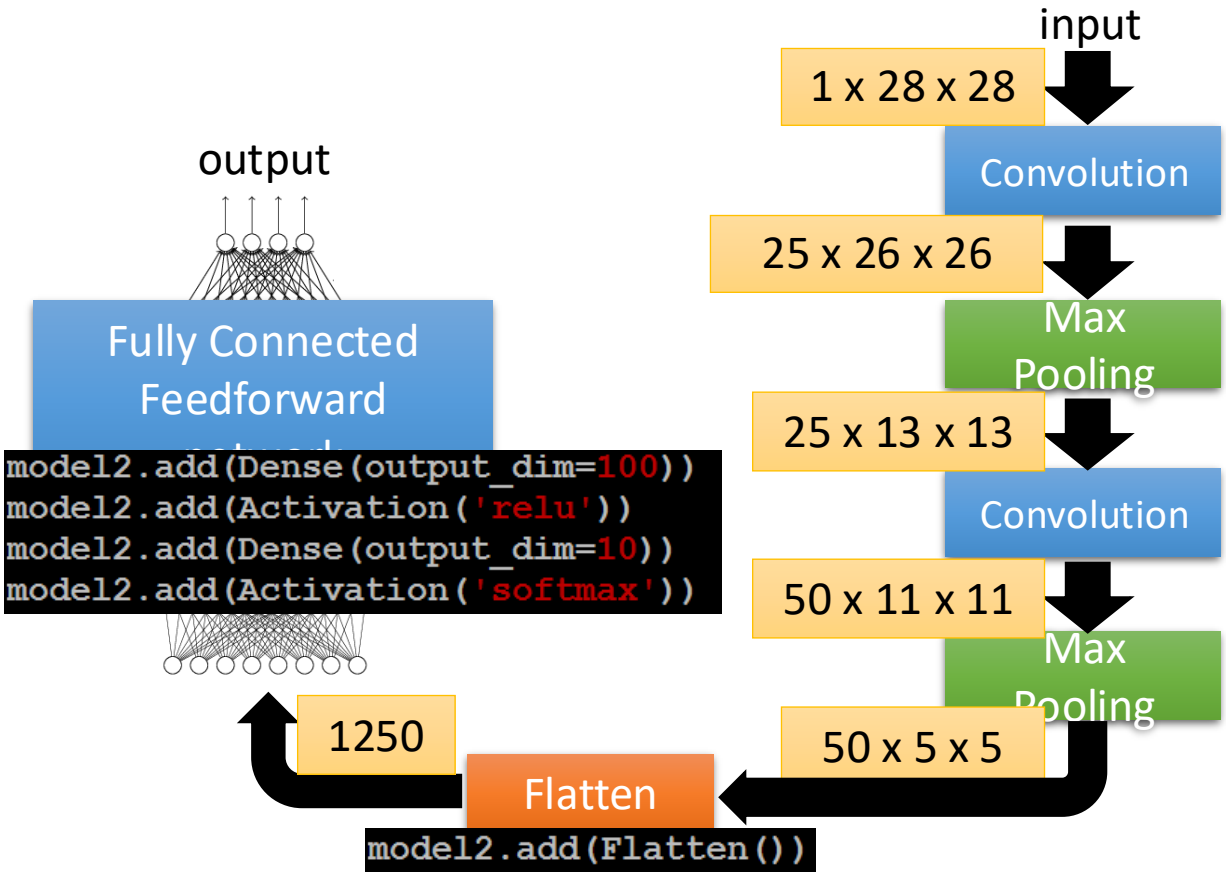
Dr. Hung-yi Lee's CNN slides

CNN in Keras



Dr. Hung-yi Lee's CNN slides

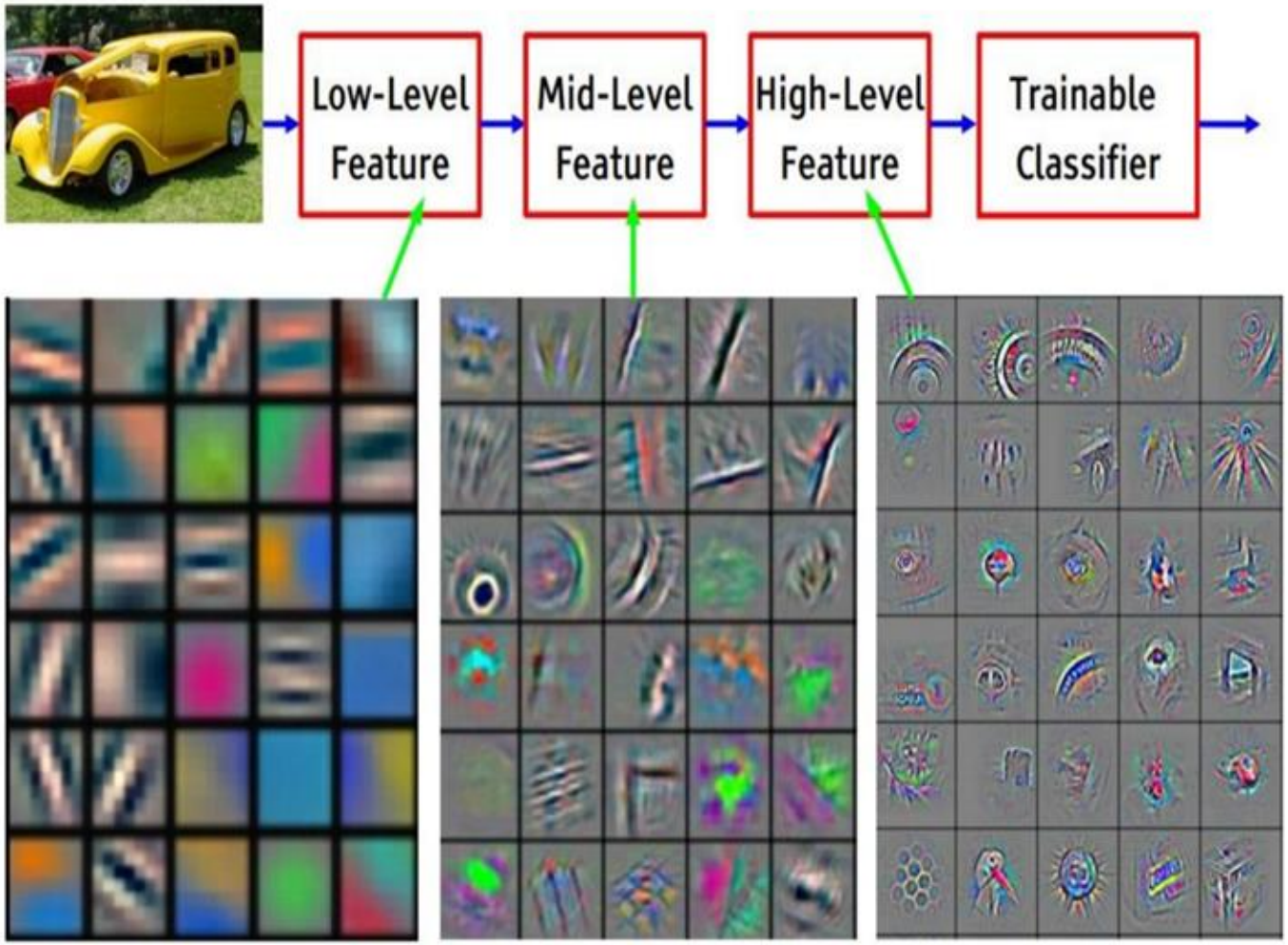
CNN in Keras



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Convolutional Neural Networks

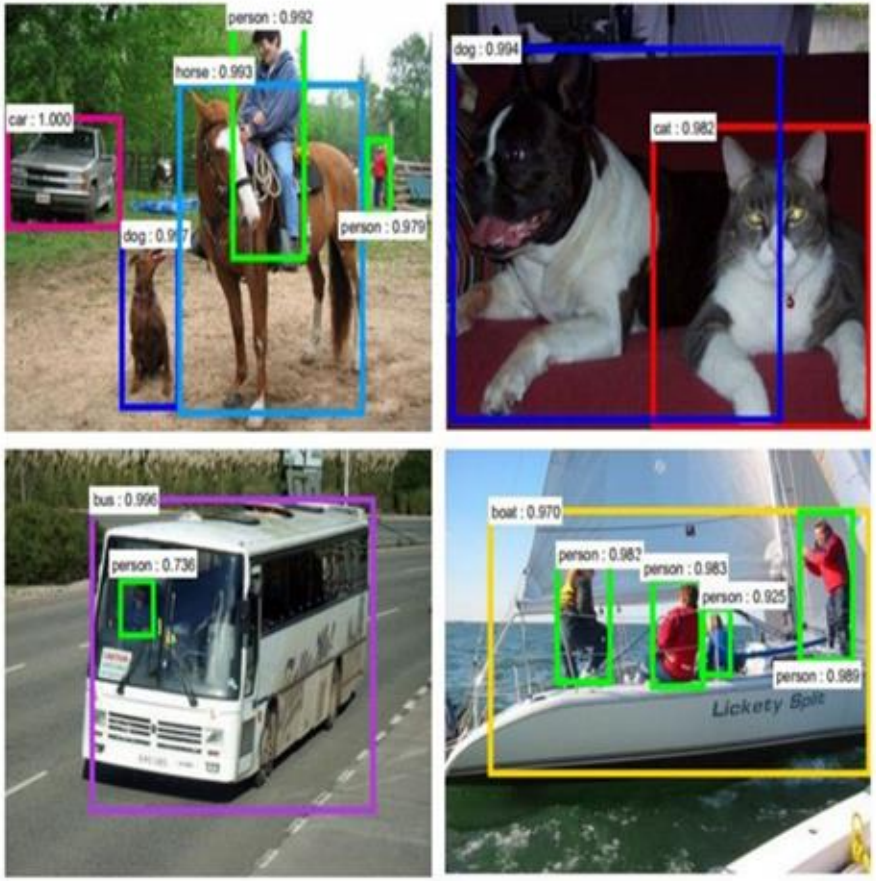
[From recent Yann LeCun slides]



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

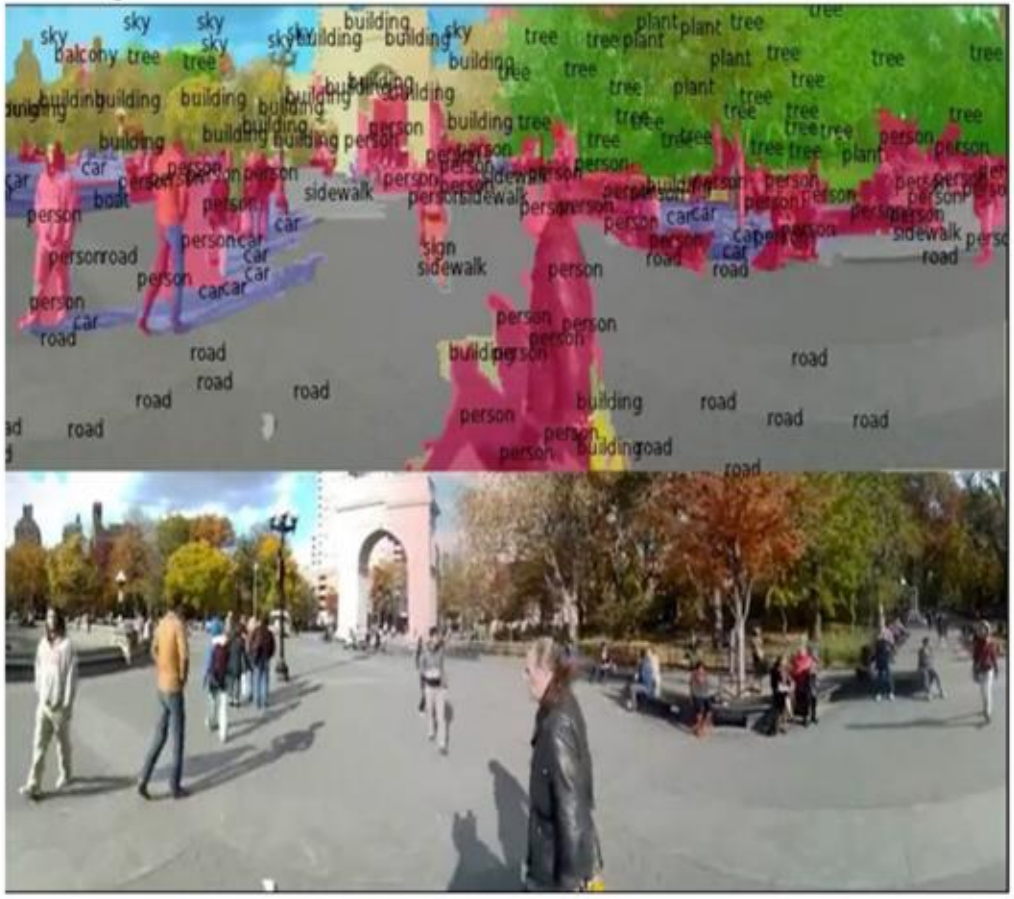
ConvNets are everywhere

Detection



[Faster R-CNN: Ren, He, Girshick, Sun 2015]

Segmentation



[Farabet et al., 2012]

More Application: Playing Go



19 x 19 matrix (image)

Network

Next move (19 x 19 positions)

19 x 19 vector

19 x 19 vector

Black: 1

white: -1

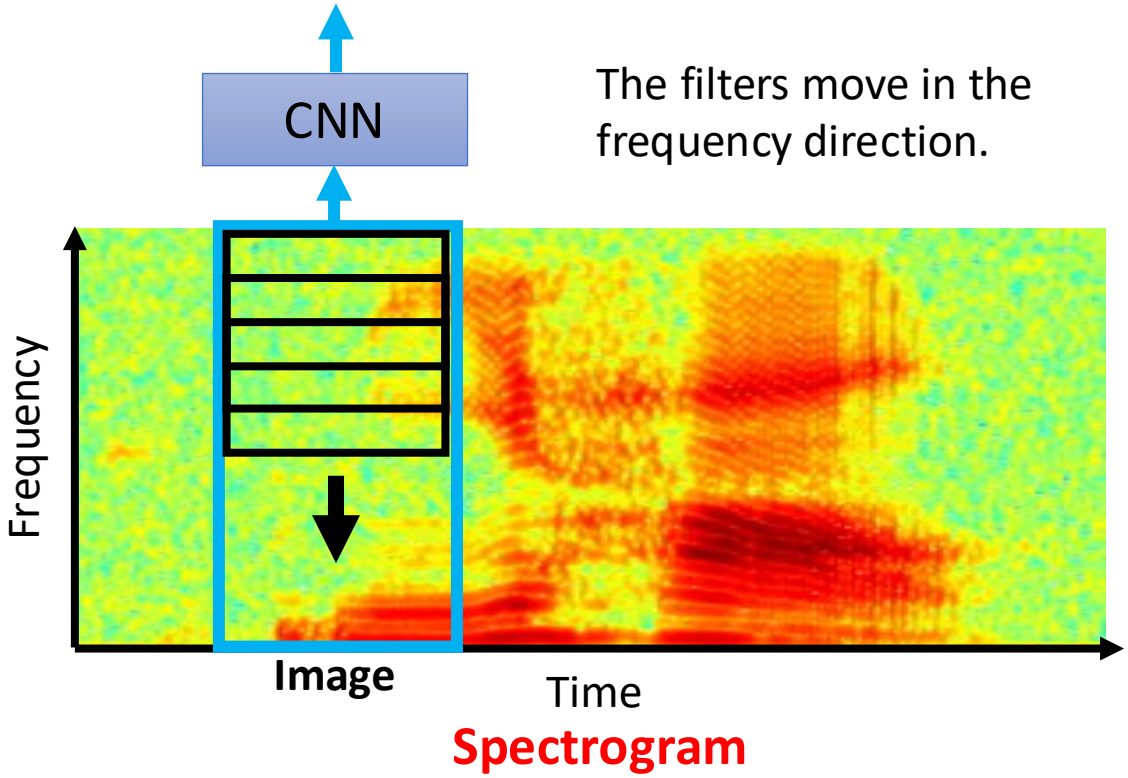
none: 0

Fully-connected feedforward network can be used

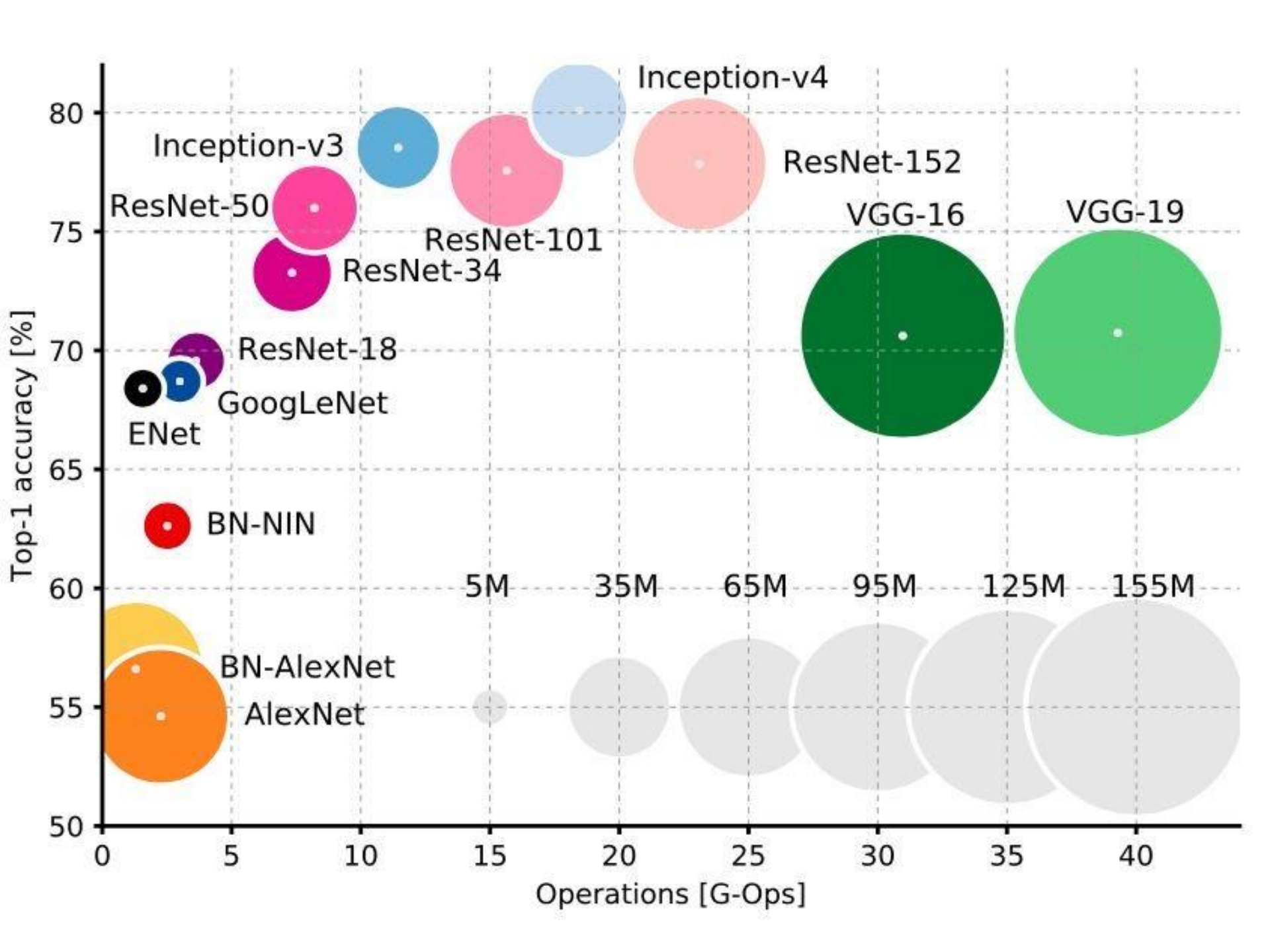
But CNN performs much better.

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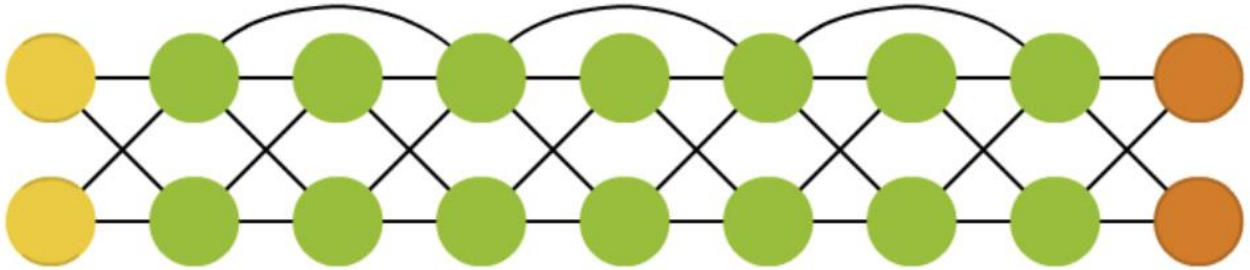
More Application: Speech



Dr. Hung-yi Lee's CNN slides


Inception-v4
Inception-v3
ResNet-152
ResNet-50
ResNet-101
VGG-16
VGG-19
ResNet-34
ResNet-18
GoogLeNet
ENet
BN-NIN
5M
35M
65M
95M
125M
155M
BN-AlexNet
AlexNet
Top-1 accuracy [%]
Operations [G-Ops]
80
75
70
65
60
55
50
0
5
10
15
20
25
30
35
40

Deep Residual Network (DRN)



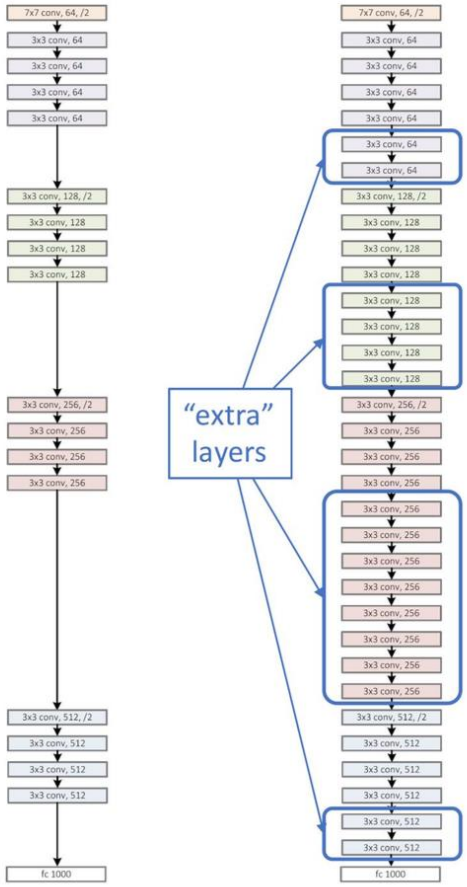
Capsule Network (CN)



Residual Trick:

Residual/Skip Connections

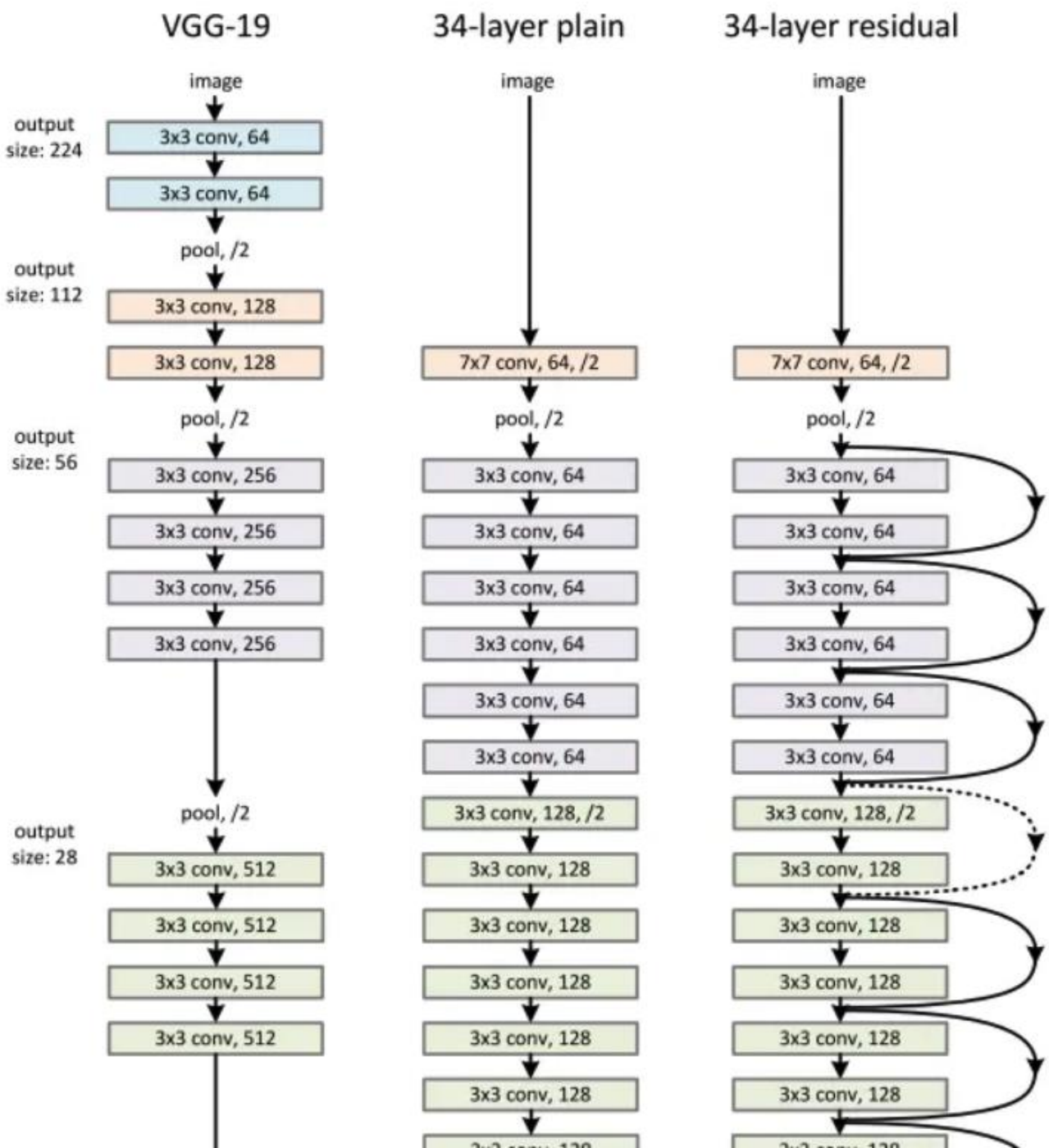
a shallower model (18 layers)

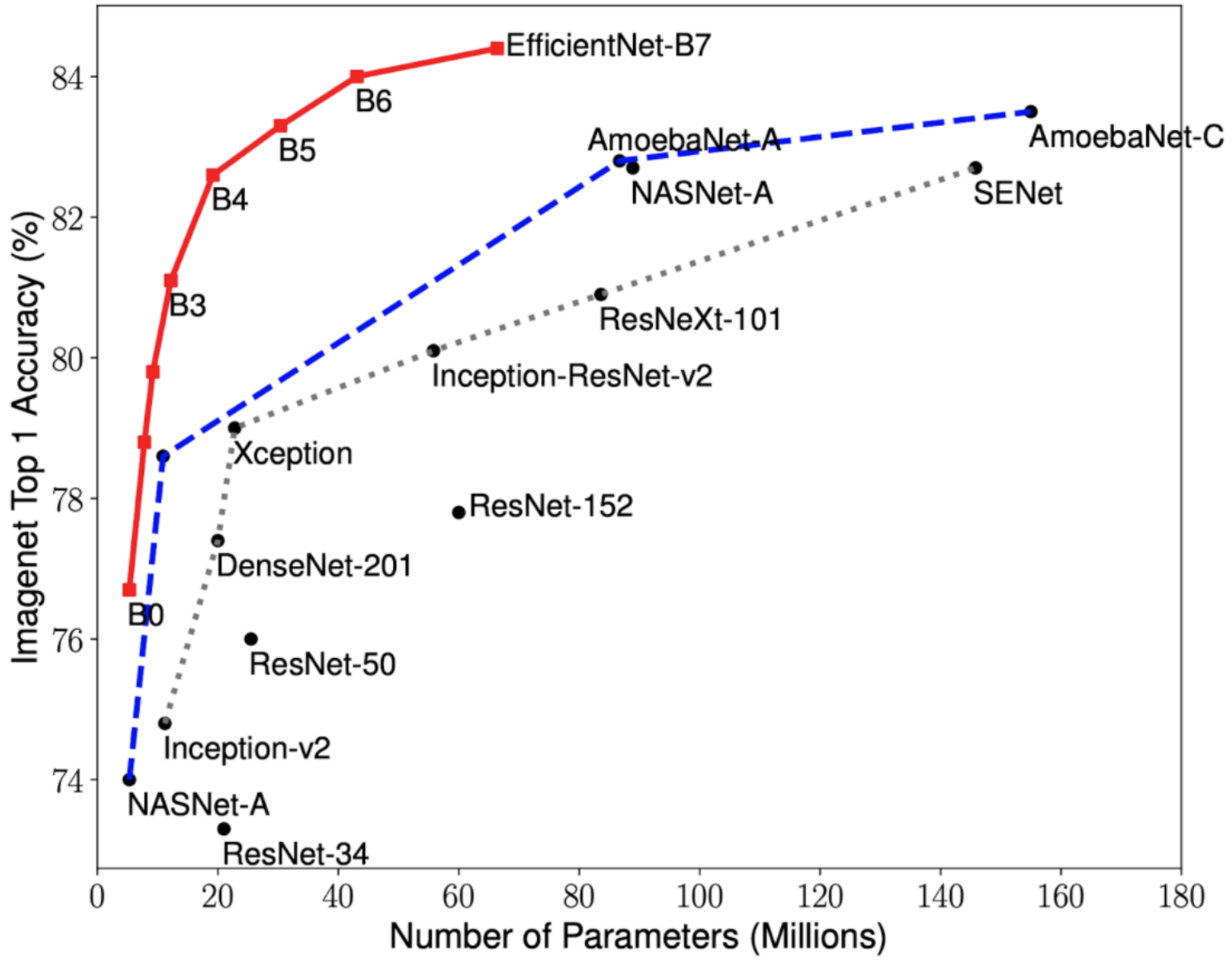


a deeper counterpart (34 layers)

- Richer solution space
- A deeper model should not have **higher training error**
- A solution *by construction*:
  - original layers: copied from a learned shallower model
  - extra layers: set as identity
  - at least the same training error
- Optimization difficulties: solvers cannot find the solution when going deeper…

Kaiming He, Xiangyu Zhang, Shaoqing Ren, & Jian Sun. "Deep Residual Learning for Image Recognition". CVPR 2016.


VGG-19
34-layer plain
34-layer residual
image
output size: 224
3x3 conv, 64
3x3 conv, 64
pool, /2
output size: 112
3x3 conv, 128
3x3 conv, 128
pool, /2
output size: 56
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
pool, /2
output size: 28
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
7x7 conv, 64, /2
pool, /2
3x3 conv, 64
3x3 conv, 128, /2
3x3 conv, 128



https://ai.googleblog.com/2019/05/efficientnet-improving-accuracy-and.html


Thank You

References

- Big thanks to Prof. Ziv Bar-Joseph and Prof. Eric Xing @ CMU for allowing me to reuse some of his slides
- Elements of Statistical Learning, by Hastie, Tibshirani and Friedman
- Prof. Andrew Moore @ CMU's slides
- Tutorial slides from Dr. Tie-Yan Liu, MSR Asia

(number of) false positive (FP)

eqv. with false alarm, Type I error

(number of) false negative (FN)

eqv. with miss, Type II error

---

sensitivity or true positive rate (TPR)

eqv. with hit rate, recall

$$TPR = TP/P = TP/(TP + FN)$$

specificity (SPC) or true negative rate

$$SPC = TN/N = TN/(TN + FP)$$

precision or positive predictive value (PPV)

$$PPV = TP/(TP + FP)$$

negative predictive value (NPV)

$$NPV = TN/(TN + FN)$$

fall-out or false positive rate (FPR)

$$FPR = FP/N = FP/(FP + TN) = 1 - SPC$$

false negative rate (FNR)

$$FNR = FN/(TP + FN) = 1 - TPR$$

false discovery rate (FDR)

$$FDR = FP/(TP + FP) = 1 - PPV$$

---

accuracy (ACC)

$$ACC = (TP + TN)/(TP + FP + FN + TN)$$

F1 score

is the harmonic mean of precision and sensitivity

$$F1 = 2TP/(2TP + FP + FN)$$

Actual Positive

PP: Predicted Positive

Actual Negative

Recall

From Wiki

When with Unbalanced Issue (binary case)

Acc Bad

#AP << #AN

- Class imbalance issue

- Balanced accuracy:

	actual	
	+	−
predicted+	*TP*	*FP*
predicted−	*FN*	*TN*

When with Unbalanced Issue (binary case)

- Class imbalance issue

num AP << num AN

actual positive actual neg.

- Balanced accuracy:

$$= \frac{1}{2}\left(\frac{TP}{PP} + \frac{TN}{PN}\right)$$

PP: TP+FP PN: TN+FN

	actual	
	+	−
predicted+	TP	FP
predicted−	FN	TN

AP vs. AN = 1 : 99

① classifier[1]

		y AP	y AN
$\hat{y}$	PP	0	0
$\hat{y}$	PN	1	99

$$ACC = \frac{99}{100} = 99\%$$

$$BACC = \frac{1}{2}\left(\frac{0}{0+\varepsilon} + \frac{99}{100}\right)$$

$$= 49.5\%$$

Low Ratio of Positive Class (binary case)

If $\frac{\text{Actual P}}{AP + AN}$ very small (e.g. < 1%)

(1, 99)
pos neg

⇒ a classifier can predict every example as Neg

⇒ ①

	AP ①	AN 99
predict P	0	0
predict N	1	99

⇒ Accuracy = $\frac{99}{100}$ = 0.99

⇒ Balanced Acc =

Bad-neg-classifier

① Balanced Acc $= \frac{1}{2}\left(\frac{TP}{P} + \frac{TN}{N}\right)$

$= \frac{1}{2}\left(\frac{0}{0+\varepsilon} + \frac{99}{100}\right) = 0.495$

$[0, \frac{1}{2}]$

another classifier

②

	AP	AN
PP	1	0
PN	0	99

Balanced Acc $= \frac{1}{2}\left(\frac{1}{1} + \frac{99}{99}\right) = 1$

$Acc = \frac{1+99}{1+0+99+0} = 1$

(posRatio 1/100)

③ Third classifier

	AP	AN
PP+	0	1
PN-	1	99

$$ACC = \frac{99}{101} \approx 99\%$$

$$BACC = \frac{1}{2}\left(\frac{0}{1} + \frac{99}{100}\right) \approx 0.495$$

(pos Ratio 2/120)

④ Fourth case

	AP	AN
PP+	1	19
PN-	1	99

$$ACC = \frac{100}{120} \approx 83\%$$

$$BACC = \frac{1}{2}\left(\frac{1}{20} + \frac{99}{100}\right) \approx 0.52$$

When with Unbalanced Issue (binary case)

4th case on previous page

another case: 2 vs. 118 ≈ 1:60

⇒ Balanced Acc cares all classes

⇒ If care more about pos+ class

	AP	AN
PP	1	19
PN	1	99

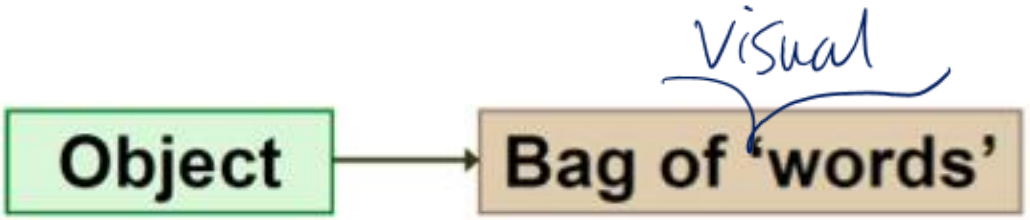
⟹ • Precision = $\frac{TP}{TP+FP}$ = $\frac{1}{20}$ = 5%

	actual +	actual −
predicted+	TP	FP
predicted−	FN	TN

• Recall = $\frac{TP}{TP+FN} = \frac{1}{2}$ = 50%

When not using Deep Learning: Image Representation for – Objective recognition

- Image representation ➔ bag of “visual words”



- An object image: histogram of visual vocabulary – a numerical vector of D dimensions.





ICML '06 Proceedings of the 23rd international conference on Machine learning

A study comparing Classifiers

An Empirical Comparison of Supervised Learning Algorithms

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Abstract

A number of supervised learning methods have been introduced in the last decade. Unfortunately, the last comprehensive empirical evaluation of supervised learning was the Statlog Project in the early 90's. We present a large-scale empirical comparison between ten supervised learning methods: SVMs, neural nets, logistic regression, naive bayes, memory-based learning, random forests, decision trees, bagged trees, boosted trees, and boosted stumps. We also examine the effect that calibrating the models via Platt Scaling and Isotonic Regression has on their performance. An important aspect of our study is the use of a variety of performance criteria to evaluate the learning methods.

This paper presents results of a large-scale empirical comparison of ten supervised learning algorithms using eight performance criteria. We evaluate the performance of SVMs, neural nets, logistic regression, naive bayes, memory-based learning, random forests, decision trees, bagged trees, boosted trees, and boosted stumps on eleven binary classification problems using a variety of performance metrics: accuracy, F-score, Lift, ROC Area, average precision, precision/recall break-even point, squared error, and cross-entropy. For each algorithm we examine common variations, and thoroughly explore the space of parameters. For example, we compare ten decision tree styles, neural nets of many sizes, SVMs with many kernels, etc.

Because some of the performance metrics we examine interpret model predictions as probabilities and models such as SVMs are not designed to predict probabil-

A study comparing Classifiers

➔ 11 binary classification problems / 8 metrics

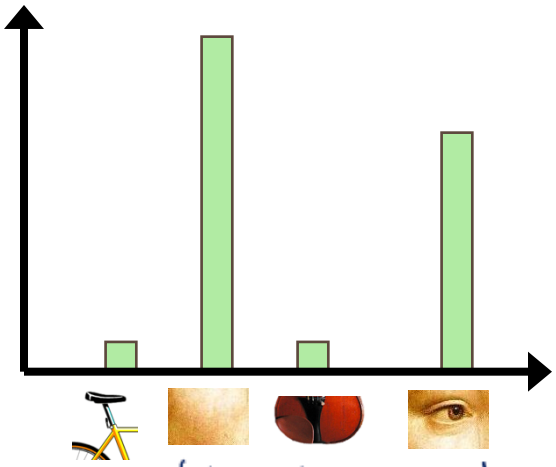
Tree, SVM, NN, DeepLearn

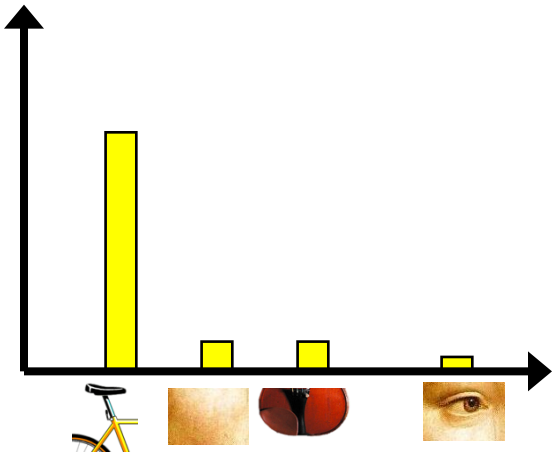
Table 2. Normalized scores for each learning algorithm by metric (average over eleven problems)

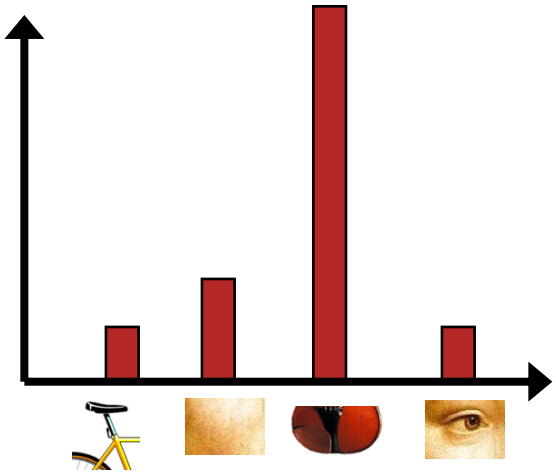
MODEL	CAL	ACC	FSC	LFT	ROC	APR	BEP	RMS	MXE	MEAN	OPT-SEL
BST-DT	PLT	.843*	.779	**.939**	**.963**	**.938**	.929*	**.880**	**.896**	**.896**	**.917**
RF	PLT	.872*	.805	.934*	.957	.931	**.930**	.851	.858	.892	.898
BAG-DT	–	.846	.781	.938*	.962*	.937*	.918	.845	.872	.887*	.899
BST-DT	ISO	.826*	.860*	.929*	.952	.921	.925*	.854	.815	.885	.917*
RF	–	**.872**	.790	.934*	.957	.931	**.930**	.829	.830	.884	.890
BAG-DT	PLT	.841	.774	.938*	.962*	.937*	.918	.836	.852	.882	.895
RF	ISO	.861*	**.861**	.923	.946	.910	.925	.836	.776	.880	.895
BAG-DT	ISO	.826	.843*	.933*	.954	.921	.915	.832	.791	.877	.894
SVM	PLT	.824	.760	.895	.938	.898	.913	.831	.836	.862	.880
ANN	–	.803	.762	.910	.936	.892	.899	.811	.821	.854	.885
SVM	ISO	.813	.836*	.892	.925	.882	.911	.814	.744	.852	.882
ANN	PLT	.815	.748	.910	.936	.892	.899	.783	.785	.846	.875
ANN	ISO	.803	.836	.908	.924	.876	.891	.777	.718	.842	.884
BST-DT	–	.834*	.816	**.939**	**.963**	**.938**	.929*	.598	.605	.828	.851
KNN	PLT	.757	.707	.889	.918	.872	.872	.742	.764	.815	.837
KNN	–	.756	.728	.889	.918	.872	.872	.729	.718	.810	.830
KNN	ISO	.755	.758	.882	.907	.854	.869	.738	.706	.809	.844
BST-STMP	PLT	.724	.651	.876	.908	.853	.845	.716	.754	.791	.808
SVM	–	.817	.804	.895	.938	.899	.913	.514	.467	.781	.810
BST-STMP	ISO	.709	.744	.873	.899	.835	.840	.695	.646	.780	.810
BST-STMP	–	.741	.684	.876	.908	.853	.845	.394	.382	.710	.726
DT	ISO	.648	.654	.818	.838	.756	.778	.590	.589	.709	.774



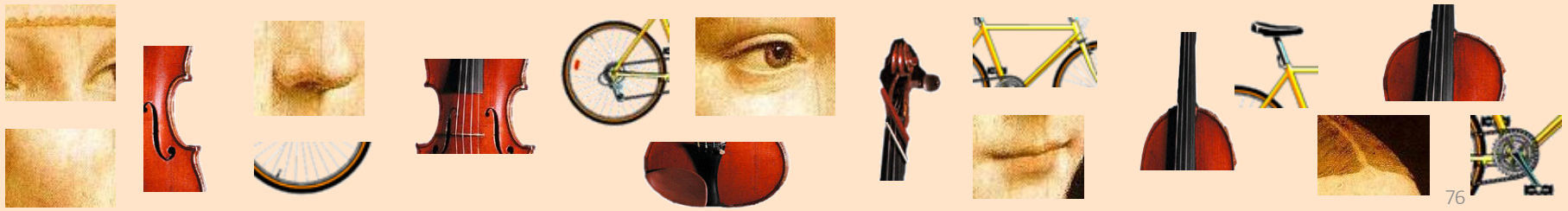
Occluded







w_1 w_2 w_3 .. w_D



A study comparing Classifiers

➔ 11 binary classification datasets

Small data

Table 1. Description of problems

PROBLEM	#ATTR	TRAIN SIZE	TEST SIZE	%POZ
ADULT	14/104	5000	35222	25%
BACT	11/170	5000	34262	69%
COD	15/60	5000	14000	50%
CALHOUS	9	5000	14640	52%
COV_TYPE	54	5000	25000	36%
HS	200	5000	4366	24%
LETTER.P1	16	5000	14000	3%
LETTER.P2	16	5000	14000	53%
MEDIS	63	5000	8199	11%
MG	124	5000	12807	17%
SLAC	59	5000	25000	50%